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Bachelor Thesis

The Ionization Problem of Atoms

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Preface

We consider an atom with the nuclear charge Z. It is a classical result of Zhislin that a nucleus can bind N electrons if $N < Z + 1$. In particular, if $Z \in \mathbb{N}$, then it can bind $N \leq Z$ electrons. Another question is: How many additional electrons can it bind? An equivalent problem is, given that this nucleus can bind some N electrons, to relate these two quantities N and Z. Empirically, one has $N \leq Z + 1$ (or, at most, $N \leq Z + 2$) [9]. It is conjectured (for example, by Barry Simon in [11], Problem 9) that this bound can rigorously be derived using the quantum-mechanical framework.

The first linear bound had been proven by E. Lieb [7] (1983)

$$N < 2Z + 1.$$

Nearly 30 years later, it had further been improved by P.T. Nam [8] (2011)

$$N < 1.22Z + 3Z^{1/3}.$$

Another decade later, that method had been refined and extended in 2025 by Hundertmark, Pattakos, Schulz [5]

$$N < 1.1185Z + O(Z^{1/3}).$$

In this thesis, we give a rigorous overview of these developments and prove several original results, namely

- Asymptotic and geometric analysis of $\alpha_{N,s}$: a quantity introduced in [5] and used to obtain bounds on the maximal excess charge;
- Proof that the symmetrization argument underlying Lemma 4.3 of [5] indeed fails for $s > 3$.

In the first chapter, we describe the mathematical model. Then we compare the quantum-mechanical formulation with the classical/geometric approach and prove a necessary condition for classical binding as well as the asymptotic neutrality as in [3]. The rest is dedicated to developing methods for obtaining necessary conditions of the form $N < CZ + o(Z)$ in the Schrödinger model. Each chapter will provide a better (tighter or easier to work with) necessary condition, eventually arriving at the asymptotically best one [5].

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Chapter 1

Introduction

1.1 Units & Notation

We use the units $\hbar = 2m_e = c = e = 1$. Thus, the momentum operator becomes

$$P = -i\nabla.$$

The atom is modeled with a nucleus of charge Z fixed at the origin. The quantum Hamiltonian acting on the antisymmetric space $\mathcal{H}_N := \bigwedge_{k=1}^{N} L^2(\mathbb{R}^3)$ is given by

$$H_{N,Z} = \sum_{j=1}^{N} P_j^2 - \sum_{j=1}^{N} \frac{Z}{|x_j|} + \sum_{1\leq j<l\leq N} \frac{1}{|x_j - x_l|}.$$

The *ground state energy* is

$$E = E_{N,Z} := \inf \sigma(H_{N,Z}).$$

1.2 Spectrum and the Binding Condition

Now, we mathematically formalize the binding condition. At first, we observe that, by the HVZ theorem,

$$\sigma_{\text{ess}}(H_{N,Z}) = [E_{N-1,Z}, \infty).$$

It implies

$$E_{N,Z} \leq E_{N-1,Z}.$$

If one has equality, it means that the system with N electrons is unstable in the sense that one electron can be removed to infinity *without paying any positive energy*. Thus, the system is stable—or *bound*—if

$$E_{N,Z} - E_{N-1,Z} < 0. \tag{1.1}$$

Note that, by the spectral theorem, $E_{N,Z}$ is an *isolated eigenvalue*, i.e. there exists a (normalized) eigenfunction $\psi_{N,Z} \in \mathcal{H}_N$ with $H_{N,Z}\psi_{N,Z} = E_{N,Z}\psi_{N,Z}$.

1.3 Motivating the Ionization Conjecture

We analyze the behavior of the eigenvector by utilizing the inequality (1.1). Consider the Hamiltonian with some (say, the k-th) electron removed:

$$H_{N,Z}^{(k)} := \sum_{\substack{j=1 \\ j\neq k}}^{N} P_j^2 - \sum_{\substack{j=1 \\ j\neq k}}^{N} \frac{Z}{|x_j|} + \sum_{\substack{1\leq j<l\leq N \\ j,l\neq k}} \frac{1}{|x_j - x_l|}.$$

Then

$$H_{N,Z} = H_{N,Z}^{(k)} + \underbrace{P_k^2 - \frac{Z}{|x_k|} + \sum_{\substack{j=1 \\ j\neq k}}^{N} \frac{1}{|x_j - x_k|}}_{k\text{-th electron's contribution}}.$$

By the variational principle, one has

$$E_{N,Z} = \inf_{\|\psi\|=1} \langle \psi, H_{N,Z}\psi \rangle ,$$

thus, by considering $H_{N,Z}^{(k)}$ as its restriction to $\mathcal{H}_{N-1}$, we have

$$\langle \phi, E_{N-1,Z}\phi \rangle = E_{N-1,Z}\|\phi\|^2 \leq \left\langle \phi, H_{N,Z}^{(k)}\phi \right\rangle \quad \text{for any admissible } \phi. \tag{1.2}$$

Using the binding condition (1.1), the inequality (1.2), and the definition of $E_{N,Z}$, one has

$$\begin{aligned} 0 < E_{N-1,Z} - E_{N,Z} &\leq \left\langle \psi_{N,Z}, H_{N,Z}^{(k)}\psi_{N,Z} \right\rangle - \langle \psi_{N,Z}, H_{N,Z}\psi_{N,Z} \rangle \\ &= \left\langle \psi_{N,Z}, \left(H_{N,Z}^{(k)} - H_{N,Z}\right)\psi_{N,Z} \right\rangle . \end{aligned}$$

By definition, the operator above is minus the k-th electron's contribution:

$$H_{N,Z}^{(k)} - H_{N,Z} = -P_k^2 + \frac{Z}{|x_k|} - \sum_{j\neq k} \frac{1}{|x_j - x_k|}.$$

Thus

$$\left\langle \psi_{N,Z}, \left(H_{N,Z}^{(k)} - H_{N,Z}\right)\psi_{N,Z} \right\rangle = \left\langle -P_k^2 \right\rangle_{\psi_{N,Z}} + \left\langle \frac{Z}{|x_k|} - \sum_{j\neq k} \frac{1}{|x_j - x_k|} \right\rangle_{\psi_{N,Z}}. \tag{1.3}$$

With integration by parts, we see that the expectation value of the negative kinetic energy is negative for any $\psi \not\equiv 0$:

$$\left\langle -P_k^2 \right\rangle_{\psi} = \langle -(-\Delta) \rangle_{\psi} = -\langle \nabla\psi_{N,Z}, \nabla\psi_{N,Z} \rangle = -\|\nabla\psi\|^2 < 0.$$

That means that if, for some k, the $\psi_{N,Z}$-expectation value of the operator $[Z/|x_k| - \sum_{j\neq k} 1/|x_j - x_k|]$ is non-positive, then the RHS of (1.3) is negative meaning that no binding is possible for these Z and N.

Now we can provide heuristics to *motivate* the ionization conjecture. Namely, if the k-th electron is being moved to infinity while the others remain bounded, we get approximately $|x_k| \approx |x_j - x_k|$ for $j \neq k$, hence

$$\frac{Z}{|x_k|} - \sum_{j \neq k} \frac{1}{|x_j - x_k|} \approx \frac{Z}{|x_k|} - \sum_{j \neq k} \frac{1}{|x_k|} = \frac{1}{|x_k|}(Z - (N-1)),$$

which is non-positive (and thus no binding should be possible) if $N \geq Z + 1$. This motivates the following formulation of THE IONIZATION CONJECTURE:

Conjecture 1.1. If an atom with nuclear charge Z can bind N electrons, then $N \leq Z + 1$, or, at most, $Z \leq N + 2$.

Chapter 2

Classical/Geometric Approach

If we admit the classical viewpoint and consider electrons as classical points in $\mathbb{R}^3$, we can reformulate the binding condition $E_{N,Z} - E_{N-1,Z} = (\text{any electron's contribution}) < 0$ in the classical sense.

2.1 Classical Binding Condition

The classical energy contribution of any (say, the k-th) electron is

$$-\frac{Z}{|x_k|} + \sum_{\substack{j=1 \\ j\neq k}}^{N} \frac{1}{|x_j - x_k|}.$$

Classically, we say that if binding occurs, then there exists some configuration of points $\{x_j\} \subset \mathbb{R}^3$ such that the energy contribution of each electron is negative, and thus, the system is stable:

$$-\frac{Z}{|x_k|} + \sum_{\substack{j=1 \\ j\neq k}}^{N} \frac{1}{|x_j - x_k|} < 0 \quad \text{for all } 1 \leq k \leq N.$$

Note that this condition is invariant under scaling all points by some constant $\lambda > 0$. Moreover, such scaling allows for each electron's energy contribution to be arbitrary small since, for all $1 \leq k \leq N$,

$$-\frac{Z}{|\lambda x_k|} + \sum_{\substack{j=1 \\ j\neq k}}^{N} \frac{1}{|\lambda x_j - \lambda x_k|} = \frac{1}{\lambda}\left[-\frac{Z}{|x_k|} + \sum_{\substack{j=1 \\ j\neq k}}^{N} \frac{1}{|x_j - x_k|}\right] \to -\infty \quad \text{as } \lambda \to 0.$$

By choosing $\lambda^{-1} := \max_j |x_j|$, we see that we can without loss of generality assume $|x_j - x_k| \leq 2$ for all pairs j, k. One then easily obtains that if $N \geq 2Z + 1$, then, for $k = \arg\max_j |x_j|$,

$$-Z + \sum_{\substack{j=1 \\ j\neq k}}^{N} \frac{1}{|x_j - x_k|} \geq -Z + \frac{N-1}{2} \geq -Z + \frac{2Z+1-1}{2} = 0,$$

hence no classical binding is possible. This proves

Theorem 2.1 (Classical Binding)**.** *If the system is bound in the classical sense, then* $N < 2Z + 1$.

2.2 Classical Asymptotic Neutrality

Empirical observations, heuristics, and the ionization conjecture suggest that atoms are asymptotically neutral, i.e. $\lim_{N\to\infty} N(Z)/Z = 1$. The following result due to Lieb, Sigal, Simon and Thirring [3] shows this in the classical setting.

Proposition 2.2. *For every $\varepsilon > 0$, there exists $N_\varepsilon > 0$ such that if $N \geq N_\varepsilon$, then for all $\{x_1, ..., x_N\} \subset \mathbb{R}^3$, one has*

$$\min_{1\leq k\leq N} \left\{ \frac{N(1-\varepsilon)}{|x_k|} - \sum_{\substack{j=1 \\ j\neq k}}^{N} \frac{1}{|x_j - x_k|} \right\} \leq 0.$$

Heuristically (cf. [9]), it means that if $N \geq \frac{1}{1-\varepsilon} Z$, then there is no binding as the previous subsection suggests.

The proof requires an auxiliary result that we will not prove here since it is a bit more involved. It can be found in [3].

Lemma 2.3 (Special case of Theorem 2.1 in [3])**.** *If ρ is a probability measure on $\mathbb{R}^3$ which is not a point mass (i.e. $\rho \neq \delta_0$), then, for any $\varepsilon > 0$, the set*

$$\left\{ y \in \mathbb{R}^3 : \int \frac{1}{|x-y|} \, \mathrm{d}\rho(x) \geq \frac{1-\varepsilon}{|y|} \right\}$$

has a positive ρ measure.

Proof of 2.2. Assume the contrary, i.e., there is some $\varepsilon_0 > 0$, a sequence $\{N_n\} \in \mathbb{N}$ with $N_n \to \infty$ as $n \to \infty$ such that

$$\frac{N_n(1-\varepsilon)}{|x_k^{(n)}|} > \sum_{j\neq k} \frac{1}{|x_j^{(n)} - x_k^{(n)}|}. \tag{2.1}$$

As we have seen above, we can without loss of generality assume $x_1^{(n)} = e_1$ and $|x_j^{(n)}| \leq 1$ for all n, j.

Define a sequence of measures

$$\rho^{(n)} := \frac{1}{N_n} \sum_{j=1}^{N_n} \delta_0(\cdot - x_j^{(n)}).$$

Note that $\rho^{(n)}$ is a sequence of probability measures supported on the unit ball. Thus the sequence is tight. Then, by the Prokhorov's theorem (cf. Theorem 5.1 in [2]), we can extract a subsequence (we call it $\{\rho_n\}$) that converges weakly $\rho_n \rightharpoonup \rho$ as $n \to \infty$ to some probability measure ρ.

Assume $y \in \overline{B_1(0)}$ is a limit point of $x_{i_n}^{(n)}$. For some $M > 0$, put $g(z) := 1/\sqrt{|z|^2 + M^2}$. Note that g is Lipschitz-continuous. Then

$$\lim_{n\to\infty} \int g(x - x_{i_n}^{(n)}) \, \mathrm{d}\rho_n(x) = \int g(x-y) \, \mathrm{d}\rho(x). \tag{2.2}$$

Note that $g(z) \leq 1/|z|$ for all $z \neq 0$. Then, using the definition of ρ_n and (2.1),

$$\int g(x - x_{i_n}^{(n)})\, \mathrm{d}\rho_n(x) < \frac{1-\varepsilon_0}{|x_{i_n}^{(n)}|}.$$

Taking limits of both sides and using (2.2) yields

$$\int g(x-y)\, \mathrm{d}\rho(x) \le \frac{1-\varepsilon_0}{|y|}.$$

By the monotone convergence theorem, we can take $M \to 0$ and obtain

$$\int \frac{1}{|x-y|}\, \mathrm{d}\rho(x) \le \frac{1-\varepsilon_0}{|y|} < \frac{1-\varepsilon}{|y|}, \quad 0 < \varepsilon < \varepsilon_0. \tag{2.3}$$

This holds for any limit point y of $x_{i_n}^{(n)}$, and thus for all $y \in \operatorname{supp}(\rho)$. Since $x_1^{(n)} = e_1$ for all n, we have (2.3) for $y = e_1$, i.e.,

$$\int \frac{1}{|x-e_1|}\, \mathrm{d}\rho(x) \le 1-\varepsilon, \quad 0 < \varepsilon < \varepsilon_0.$$

This shows $\rho \neq \delta_0$ (otherwise the left-hand side would be 1). Since (2.3) holds for all $y \in \operatorname{supp}(\rho)$, we get a contradiction to Lemma 2.3. □

Chapter 3

Quantum/PDE Method

For the rest of the thesis, we focus on the quantum-mechanical model. The first uniform bound has been obtained by E. Lieb [7] (1983) and turns out to also be $N < 2Z + 1$. We rigorously prove it in this chapter.

Let's assume the binding condition

$$E_{N,Z} < E_{N-1,Z}.$$

Let $\psi := \psi_{N,Z}$ be (the normalized $\|\psi\|_{L^2} = 1$) eigenfunction of $H_{N,Z}$ to $E_{N,Z}$:

$$E_{N,Z}\psi = H_{N,Z}\psi = H^{(k)}_{N,Z}\psi + P_k^2\psi - \frac{Z}{|x_k|}\psi + \sum_{\substack{j=1\\ j\neq k}}^{N} \frac{1}{|x_j - x_k|}\psi.$$

Now, we take the scalar product weighted by $|x_k|$:

$$\begin{aligned}\langle |x_k|\psi, E_{N,Z}\psi\rangle = &\left\langle |x_k|\psi, H^{(k)}_{N,Z}\psi\right\rangle + \\ &+ \left\langle |x_k|\psi, P_k^2\psi\right\rangle - \underbrace{\left\langle |x_k|\psi, \frac{Z}{|x_k|}\psi\right\rangle}_{Z\langle\psi,\psi\rangle = Z} + \sum_{\substack{j=1\\ j\neq k}}^{N}\left\langle |x_k|\psi, \frac{1}{|x_j - x_k|}\psi\right\rangle. \end{aligned} \tag{3.1}$$

Consider the first term. Since $H^{(k)}_{N,Z}$ commutes with $|x_k|$ and, for any fixed k, the function $|x_k|\psi$ has the same symmetry as ψ in the other $N-1$ variables, one has

$$\begin{aligned}\left\langle |x_k|\psi, H^{(k)}_{N,Z}\psi\right\rangle &= \left\langle \left(|x_k|^{1/2}\psi\right), H^{(k)}_{N,Z}\left(|x_k|^{1/2}\psi\right)\right\rangle \in \mathbb{R} \\ &\geq E_{N-1,Z}\left\langle |x_k|\psi, \psi\right\rangle. \end{aligned} \tag{3.2}$$

By combining (3.2) and (3.1) and taking the real part, we obtain

$$\begin{aligned}\mathrm{Re}\left\langle |x_k|\psi, E_{N,Z}\psi\right\rangle &= E_{N,Z}\left\langle |x_k|\psi, \psi\right\rangle \\ &\geq E_{N-1,Z}\left\langle |x_k|\psi, \psi\right\rangle + \mathrm{Re}\left\langle |x_k|\psi, P_k^2\psi\right\rangle - Z + \sum_{\substack{j=1\\ j\neq k}}^{N}\left\langle \psi, \frac{|x_k|}{|x_j - x_k|}\psi\right\rangle. \end{aligned}$$

Equivalently,

$$Z - \sum_{\substack{j=1\\ j\neq k}}^{N}\left\langle \psi, \frac{|x_k|}{|x_j - x_k|}\psi\right\rangle \geq \underbrace{(E_{N-1,Z} - E_{N,Z})}_{>0 \text{ (binding condition!)}}\left\langle \psi, |x_k|\psi\right\rangle + \mathrm{Re}\left\langle |x_k|\psi, P_k^2\psi\right\rangle.$$

This yields the following strict inequality:

$$Z - \sum_{\substack{j=1 \\ j\neq k}}^{N} \left\langle \psi, \frac{|x_k|}{|x_j - x_k|}\psi \right\rangle > \operatorname{Re} \left\langle |x_k|\psi, P_k^2\psi \right\rangle . \tag{3.3}$$

The following lemma (due to [7]) implies that the right-hand side is positive. We will prove it later.

Lemma 3.1. *Let $f \in H^1(\mathbb{R}^3)$ satisfy $\|f\|_{L^2} = 1$, and let $\phi : \mathbb{R}^3 \to (0, \infty]$, be such that $-\Delta\phi$ is a positive measure. Assume that $f/\phi \in H^1(\mathbb{R}^3)$ (where $1/\phi = 0$ on $\{\phi = \infty\}$) and that $\{\phi = \infty\}$ is a null set. Then*

$$\operatorname{Re} \int \frac{\overline{f}}{\phi}(-\Delta)f > 0.$$

Since $-\Delta(1/|x|) = 4\pi\delta_0$ is a positive measure (the equality holds in the sense of distributions), this lemma can be applied with $\phi(x) := 1/|x|$, $f(x) = \psi(x_1, ..., x_{k-1}, x, x_{k+1}, ..., x_N)$ for some fixed $\{x_j : j \neq k\} \subset \mathbb{R}^3$. Hence the right-hand side of (3.3) is actually positive which yields

$$Z - \sum_{\substack{j=1 \\ j\neq k}}^{N} \left\langle \psi, \frac{|x_k|}{|x_j - x_k|}\psi \right\rangle > 0 \quad \text{for all } 1 \leq k \leq N$$

as a necessary condition for binding. Taking the sum of both sides over $k \in \{1, \dots, N\}$ gives

$$ZN - \sum_{k=1}^{N} \sum_{\substack{j=1 \\ j\neq k}}^{N} \left\langle \psi, \frac{|x_k|}{|x_j - x_k|}\psi \right\rangle > 0.$$

We symmetrize the sum

$$ZN - \frac{1}{2} \left\langle \psi, \underbrace{\sum_{k\neq j}}_{N(N-1) \text{ terms}} \underbrace{\frac{|x_k| + |x_j|}{|x_j - x_k|}}_{\geq 1} \psi \right\rangle > 0.$$

and obtain

$$NZ - \frac{1}{2}N(N-1) > 0$$

which proves the famous bound

$$N < 2Z + 1. \tag{3.4}$$

In particular, it implies that the doubly ionized hydrogen atom is unstable which explains the experimental data. The only thing left is

Proof of Lemma 3.1. Let $(\cdot, \cdot)$ represent the standard real scalar (dot) product on $\mathbb{R}^3$. Consider $g := f/\phi \in H^1(\mathbb{R}^3)$. Then we need to show

$$I := -\operatorname{Re} \int \frac{\overline{f}}{\phi}\Delta f = -\operatorname{Re} \int \overline{g}\Delta(g\phi) > 0.$$

Note that $\int \phi|\nabla g|^2 > 0$. Indeed, if the integral were zero, g would be an a.e.-constant function. Since $g \in H^1(\mathbb{R}^3)$ by assumption, it would force $g \equiv 0$ a.e. Hence $f \equiv 0$ a.e. (which is ruled out by the normalization) or $\phi \equiv \infty$ a.e. which contradicts the assumption.

We integrate by parts:

$$I = \mathrm{Re} \int (\nabla \overline{g}, \nabla(g\phi)).$$

Then we use the product rule together with another integration by parts to complete the proof:

$$\begin{aligned}
I = \mathrm{Re} \int (\nabla \overline{g}, \nabla(g\phi)) &= \mathrm{Re} \int (\nabla \overline{g}, (\nabla g)\phi + g(\nabla \phi)) \\
&= \mathrm{Re} \int \underbrace{(\nabla \overline{g}, \nabla g)}_{|\nabla g|^2} \phi + \mathrm{Re} \int (\nabla \overline{g}, g \nabla \phi) \\
&= \underbrace{\int \phi |\nabla g|^2}_{>0} + \frac{1}{2} \left(\int (\nabla \overline{g}, g\nabla\phi) + \overline{\int (\nabla \overline{g}, g \nabla \phi)} \right) \\
&> \frac{1}{2} \int (\nabla \phi, (\nabla \overline{g}) g) + (\nabla \phi, \overline{g} \nabla g) \\
&= \frac{1}{2} \int (\nabla \phi, \nabla(g, \overline{g})) = \frac{1}{2} \int (\nabla \phi, \nabla |g|^2) \\
&= \frac{1}{2} \int (-\Delta \phi) |g|^2 \geq 0.
\end{aligned}$$

□

This section demonstrates that $N < 2Z + 1$ is a necessary condition for binding. We derived it by assuming that $\psi_{N,Z}$ is a bound state with $E_{N,Z} < E_{N-1,Z}$. In the next section, under the same assumption, we refine the method to obtain a tighter bound on N in terms of Z.

Chapter 4

An Extension and Refinements of the PDE Method

A natural question is why one restricts oneself to the choice of the weight $|x_k|$ in the quantity $\langle |x_k|\psi_{N,Z}, H_{N,Z}\psi_{N,Z}\rangle$. It turns out that the bound (3.4) can be improved by replacing $|x_k|$ by $|x_k|^s$ for $s > 1$. This section provides an overview of this approach. In successive chapters, we present several original proofs of properties of the auxiliary tools used in the analysis.

Originally, the improvement had been achieved by P.T. Nam [8] for s=2. The method had later been extended by HPS [5] to $s > 1$, and we follow this more general formulation here.

4.1 Using a Better Weight

We proceed similarly to Section 3 (as in HPS [5]):

$$\begin{aligned}\langle |x_k|^s\psi, E_{N,Z}\psi\rangle =& \left\langle |x_k|^s\psi, H^{(k)}_{N,Z}\psi\right\rangle \\ &+ \langle |x_k|^s\psi, P_k^2\psi\rangle - \underbrace{\left\langle |x_k|^s\psi, \frac{Z}{|x_k|}\psi\right\rangle}_{Z\langle |x_k|^{s-1}\psi,\psi\rangle} + \sum_{\substack{j=1\\ j\neq k}}^{N}\left\langle |x_k|^s\psi, \frac{1}{|x_j - x_k|}\psi\right\rangle .\end{aligned}$$

Again, since $|x_k|^{s/2}$ commutes with $H^{(k)}_{N,Z}$, by the variational principle

$$\left\langle |x_k|^s\psi, H^{(k)}_{N,Z}\psi\right\rangle = \left\langle \left(|x_k|^{s/2}\psi\right), H^{(k)}_{N,Z}\left(|x_k|^{s/2}\psi\right)\right\rangle \geq E_{N-1,Z}\,\langle |x_k|^s\psi, \psi\rangle .$$

Hence

$$\begin{aligned}E_{N,Z}\,\langle |x_k|^s\psi, \psi\rangle \geq& E_{N-1,Z}\,\langle |x_k|^s\psi, \psi\rangle \\ &+ \mathrm{Re}\,\langle |x_k|^s\psi, P_k^2\psi\rangle - Z\,\langle |x_k|^{s-1}\psi, \psi\rangle + \sum_{\substack{j=1\\ j\neq k}}^{N}\left\langle \psi, \frac{|x_k|^s}{|x_j - x_k|}\psi\right\rangle .\end{aligned}$$

Equivalently,

$$Z\,\langle |x_k|^{s-1}\rangle - \left\langle \sum_{\substack{j=1\\ j\neq k}}^{N}\frac{|x_k|^s}{|x_j - x_k|}\right\rangle \geq \underbrace{(E_{N-1,Z} - E_{N,Z})}_{>0 \text{ (binding condition!)}}\,\langle \psi, |x_k|^s\psi\rangle + \mathrm{Re}\,\langle |x_k|^s\psi, P_k^2\psi\rangle .$$

Summing over $k \in \{1, \dots, N\}$ yields the following strict inequality:

$$Z \left\langle \sum_{k=1}^{N} |x_k|^{s-1} \right\rangle - \left\langle \sum_{1 \le j < k \le N} \frac{|x_k|^s + |x_j|^s}{|x_j - x_k|} \right\rangle > \sum_{k=1}^{N} \mathrm{Re} \left\langle |x_k|^s \psi, P_k^2 \psi \right\rangle .$$

Note that $\langle |x_k|^{s-1} \rangle = \langle |x_1|^{s-1} \rangle$ and $\langle |x_k|^s \psi, P_k^2 \psi \rangle = \langle |x_1|^s \psi, P_1^2 \psi \rangle$ for all k. Thus

$$\frac{\left\langle \sum_{1 \le j < k \le N} \frac{|x_k|^s + |x_j|^s}{|x_j - x_k|} \right\rangle}{\left\langle \sum_{k=1}^{N} |x_k|^{s-1} \right\rangle} < Z - \frac{\mathrm{Re} \langle |x_1|^s \psi, P_1^2 \psi \rangle}{\langle |x_1|^{s-1} \rangle}.$$

Now, we introduce a crucial tool for our analysis as it is defined in HPS [5]:

$$\alpha_{N,s} := \inf \left\{ \frac{\sum_{1 \le j < k \le N} \frac{|x_j|^s + |x_k|^s}{|x_j - x_k|}}{(N-1) \sum_{k=1}^{N} |x_k|^{s-1}} \ : \ x_1, \dots, x_N \in \mathbb{R}^3 \right\}.$$

By definition,

$$\alpha_{N,s} (N-1) \sum_{k=1}^{N} |x_k|^{s-1} \le \sum_{1 \le j < k \le N} \frac{|x_k|^s + |x_j|^s}{|x_j - x_k|}.$$

Taking the expectation values of both sides yields

$$\alpha_{N,s} (N-1) \le \frac{\left\langle \sum_{1 \le j < k \le N} \frac{|x_k|^s + |x_j|^s}{|x_j - x_k|} \right\rangle}{\left\langle \sum_{k=1}^{N} |x_k|^{s-1} \right\rangle},$$

hence

$$\alpha_{N,s} (N-1) < Z - \frac{\mathrm{Re} \langle |x_1|^s \psi, P_1^2 \psi \rangle}{\langle |x_1|^{s-1} \psi, \psi \rangle}. \tag{4.1}$$

This shows that (4.1) is necessary for $E_{N,Z} < E_{N-1,Z}$. Thus, if we have a good lower bound on $\alpha_{N,s}$ as well as enough control of the weighted kinetic energy term on the right-hand side, an inequality of the form of $N < CZ + o(Z)$ would follow.

4.2 Weighted Kinetic Energy Term

The next objective is to estimate the kinetic contribution appearing in (4.1), namely

$$- \frac{\mathrm{Re} \langle |x_1|^s \psi, P_1^2 \psi \rangle}{\langle |x_1|^{s-1} \psi, \psi \rangle}, \qquad s \ge 1. \tag{4.2}$$

The numerator is understood in the quadratic-form sense. That is,

$$\mathrm{Re} \langle |x_1|^s \psi, -\Delta_1 \psi \rangle = \mathrm{Re} \langle \nabla_1 (|x_1|^s \psi), \nabla_1 \psi \rangle,$$

where the gradient acts only on the first particle variable.

For the linear weight $s = 1$, we proved (cf. Lemma 3.1)

$$-\operatorname{Re}\langle |x|\psi, \Delta\psi\rangle \geq 0,$$

which immediately eliminates the kinetic contribution. However, this positivity is special to the case $s = 1$ and does not persist for larger exponents.

To control the weighted kinetic energy for general $s > 1$, we proceed as in Section 6 of [5]. The following lemma provides the first estimate on the weighted kinetic term.

Lemma 4.1. *For any $s \geq 2$,*

$$-\frac{\operatorname{Re}\langle |x_1|^s\psi, (-\Delta_1)\psi\rangle}{\langle |x_1|^{s-1}\psi, \psi\rangle} \leq \frac{s^2-1}{4}\left\langle |x_1|^{s-1}\psi, \psi\right\rangle^{-1/(s-1)}. \tag{4.3}$$

Proof. Assume $s > 2$. We approximate $x \mapsto |x_1|^{s/2}$ by $\xi_\varepsilon \in W^{1,\infty}(\mathbb{R}^{3N})$ defined as

$$\xi_\varepsilon(x) := \frac{|x_1|^{s/2}}{1+\varepsilon|x_1|^{s/2}} \leq |x_1|^{s/2}, \qquad x \in \mathbb{R}^{3N}.$$

Using the IMS localization formula and the Hardy inequality, (see Appendix A.3), we obtain

$$\operatorname{Re}\left\langle \xi_\varepsilon^2\psi, (-\Delta_1)\psi\right\rangle \geq \left\langle \psi, \left(\frac{\xi_\varepsilon^2}{4|x_1|^2} - |\nabla_1\xi_\varepsilon|^2\right)\psi\right\rangle.$$

Note that

$$\begin{aligned}\lim_{\varepsilon\to 0}\left(\frac{\xi_\varepsilon^2}{4|x_1|^2} - |\nabla_1\xi_\varepsilon|^2\right) &= \frac{|x_1|^s}{4|x_1|^2} - \left(\frac{s}{2}|x_1|^{\frac{s}{2}-1}\right)^2 \\ &= \frac{1}{4}|x_1|^{s-2} - \frac{s^2}{4}|x_1|^{s-2} = \frac{1-s^2}{4}\,|x_1|^{s-2}.\end{aligned}$$

Dominated convergence yields, as $\varepsilon \to 0$,

$$-\frac{\operatorname{Re}\langle |x_1|^s\psi, (-\Delta_1)\psi\rangle}{\langle |x_1|^{s-1}\psi, \psi\rangle} \leq \frac{s^2-1}{4}\,\frac{\langle |x_1|^{s-2}\psi, \psi\rangle}{\langle |x_1|^{s-1}\psi, \psi\rangle}. \tag{4.4}$$

It remains to estimate the ratio of moments. Introduce the one-particle density

$$\rho_\psi(x) = N\int_{\mathbb{R}^{3(N-1)}} |\psi(x, x_2, \ldots, x_N)|^2\,\mathrm{d}x_2\cdots\mathrm{d}x_N, \qquad \int_{\mathbb{R}^3}\rho_\psi = N.$$

Then for any exponent $p \geq 0$,

$$\langle |x_1|^p\psi, \psi\rangle = \frac{1}{N}\int_{\mathbb{R}^3} |x|^p\,\rho_\psi(x)\,\mathrm{d}x.$$

Consequently,

$$\frac{\langle |x_1|^{s-2}\psi, \psi\rangle}{\langle |x_1|^{s-1}\psi, \psi\rangle} = \frac{\int |x|^{s-2}\rho_\psi(x)\,\mathrm{d}x}{\int |x|^{s-1}\rho_\psi(x)\,\mathrm{d}x} = \frac{\int |x|^{s-2}\,\mathrm{d}\mu(x)}{\int |x|^{s-1}\,\mathrm{d}\mu(x)},$$

where $\mu(\mathrm{d}x) = \rho_\psi(x)\mathrm{d}x/\|\rho_\psi\|_{L^1}$ is a probability measure on $\mathbb{R}^3$. By Hölder's inequality,

$$\int |x|^{s-2}\,\mathrm{d}\mu \leq \left(\int |x|^{s-1}\,\mathrm{d}\mu\right)^{\frac{s-2}{s-1}}\left(\int \mathrm{d}\mu\right)^{\frac{1}{s-1}} = \left(\int |x|^{s-1}\,\mathrm{d}\mu\right)^{\frac{s-2}{s-1}}.$$

Thus

$$\frac{\int |x|^{s-2}\,\mathrm{d}\mu}{\int |x|^{s-1}\,\mathrm{d}\mu} \;\leq\; \left(\int |x|^{s-1}\,\mathrm{d}\mu\right)^{-\frac{1}{s-1}} = \left(\frac{1}{N}\int |x|^{s-1}\rho_\psi(x)\,\mathrm{d}x\right)^{-\frac{1}{s-1}} = \left\langle |x_1|^{s-1}\psi,\psi\right\rangle^{-\frac{1}{s-1}}.$$

Inserting this into (4.4) yields

$$-\frac{\mathrm{Re}\,\langle |x_1|^s\psi,(-\Delta_1)\psi\rangle}{\langle |x_1|^{s-1}\psi,\psi\rangle} \;\leq\; \frac{s^2-1}{4}\,\left\langle |x_1|^{s-1}\psi,\psi\right\rangle^{-\frac{1}{s-1}}.$$

The case $s=2$ is covered by taking $s\to 2$ (cf. Lemma 6.1 in [5]). □

Lemma 4.2. *Let $p\geq 0$. Then*

$$\langle |x_1|^p\psi,\psi\rangle^{-1/p} = \left(\frac{1}{N}\int |x|^p\rho_\psi(x)\,\mathrm{d}x\right)^{-1/p} \leq \frac{1}{2}\kappa\, C_p^{-1/p}\, Z N^{-2/3},$$

where $\kappa = \sqrt{5}\left(\frac{2}{9\pi^2}1.456\right)^{1/3}$ and C_p is the constant in the following inequality introduced by Lieb in [6]:

$$\left(\int_{\mathbb{R}^3} f(x)^{5/3}\,\mathrm{d}x\right)^{p/2}\int_{\mathbb{R}^3}|x|^p\,f(x)\,\mathrm{d}x \geq C_p\left(\int_{\mathbb{R}^3} f(x)\,\mathrm{d}x\right)^{1+(5p)/6}. \tag{4.5}$$

Remark 4.3. The sharp constant C_p is attained for $f_p(x) := \chi_{\{|y|\leq 1\}}(x)\,(1-|x|^p)^{3/2}$. This can be verified using Euler-Lagrange equations. The constant is given in terms of the Gamma function and can be found in Appendix A.5

Proof. We apply (4.5) with $f=\rho_\psi$ as in proof of Lemma 4.1. This yields

$$\left(\frac{1}{N}\int_{\mathbb{R}^3}|x_1|^p\,\rho_{\psi_{N,Z}}(x_1)\,dx_1\right)^{-1/p} \leq C_p^{-1/p}N^{-5/6}\left(\int_{\mathbb{R}^3}\rho_{\psi_{N,Z}}(x_1)^{5/3}\,dx_1\right)^{1/2}.$$

By the fermionic kinetic energy inequality (cf. [5]),

$$2^{-2/3}\,K_3\int_{\mathbb{R}^3}\rho_{\psi_{N,Z}}(x)^{5/3}\,dx \leq \sum_{j=1}^N \langle\psi_{N,Z},(-\Delta_j)\,\psi_{N,Z}\rangle\,, \tag{4.6}$$

where $K_3 = \frac{3}{5}\left(\frac{1.456}{6\pi^2}\right)^{-2/3}$. By the quantum mechanical virial theorem,

$$-\,E_{N,Z} = \sum_{j=1}^N \langle\psi_{N,Z},(-\Delta_j)\,\psi_{N,Z}\rangle\,. \tag{4.7}$$

Combining (4.6) and (4.7), we obtain

$$\int_{\mathbb{R}^3}\rho_{\psi_{N,Z}}(x)^{5/3}\,dx \leq -\frac{2^{2/3}}{K_3}E_{N,Z}.$$

Together with

$$-E_{N,Z}\leq AZ^2N^{1/3},$$

where $A = (3/2^4)^{1/3}$ [5, Lemma A.7], this yields

$$\left(\frac{1}{N}\int_{\mathbb{R}^3} |x_1|^p \rho_{\psi_{N,Z}}(x_1)\,dx_1\right)^{-1/p} \leq C_p^{-1/p}\left(\frac{2^{2/3}}{K_3}A\right)^{1/2} ZN^{-2/3}$$
$$= \frac{1}{2}C_p^{-1/p}\sqrt{5}\left(\frac{2}{9\pi^2}1.456\right)^{1/3} ZN^{-2/3}.$$

□

Combining these lemmas with $p = s - 1$, we finally arrive at

Lemma 4.4. *Let $s \geq 2$. Then*

$$\alpha_{N,s}(N-1) < Z\left[1 + \frac{s^2-1}{8}\, C_{s-1}^{-1/(s-1)}\kappa N^{-2/3}\right], \tag{4.8}$$

where $\kappa = \sqrt{5}\left(\frac{2}{9\pi^2}1.456\right)^{1/3}$.

This section shows that Equation (4.8) is a necessary condition for binding $E_{N-1,Z} > E_{N,Z}$. The next step is to find better understand $\alpha_{N,s}$ in order to find a good lower bound on it.

Chapter 5

Analysis of $\alpha_{N,s}$

This section is dedicated to understanding $\alpha_{N,s}$ purely as a geometric constant. We prove a universal lower bound, analyze its asymptotic behavior, compute some exact values, and investigate its geometric properties.

5.1 Basic Properties

Let $\mathcal{D}_N := \{X = (x_1, \ldots, x_N) \in \mathbb{R}^3 : x_i \neq x_j \text{ for } i \neq j\}$. As a shorthand, put

$$A_{N,s}(X) := \frac{\displaystyle\sum_{1 \leq j < k \leq N} \frac{|x_j|^s + |x_k|^s}{|x_j - x_k|}}{(N-1)\displaystyle\sum_{k=1}^{N} |x_k|^{s-1}}$$

for $X \in \mathcal{D}_N$.

Lemma 5.1 (Homogeneity of degree 0)**.** *Let $N \geq 2$, $s \geq 1$, $X \in \mathcal{D}_N$. Then for all $\lambda > 0$:*

$$A_{N,s}(X) = A_{N,s}(\lambda X).$$

Proof. Both numerator and denominator scale as λ^{s-1}. □

Corollary 5.2. *Without loss of generality, we may assume when investigating $A_{N,s}$ that $|x_1| = 1$. In particular, $x_1 = e_1$, where e_j is the j-th basis vector with respect to the standard basis, and $|x_j| \leq 1$ for all j.*

5.2 Numerical Analysis

Numerical experiments with $\alpha_{N,s}$ served several purposes. They suggested candidate minimizers, provided insight into the asymptotic behavior, and motivated a number of conjectures. Some of these conjectures have since been proven and will be discussed in the following subsections. To facilitate numerical simulations, a custom Python framework has been developed to help compute, organize, refine, and analyze the data. An extensive dataset has been gathered. In this subsection, we will outline the algorithm and discuss some representative numerical results.

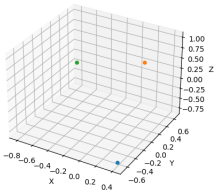
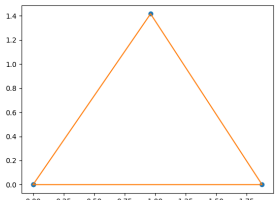
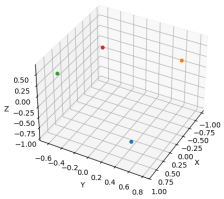
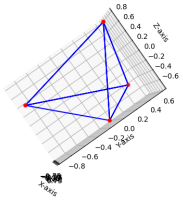
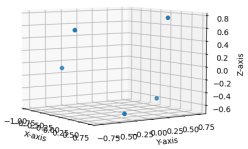
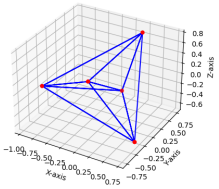
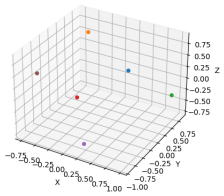
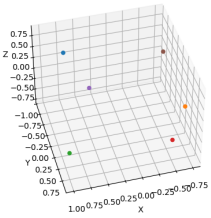
The algorithm used for approximating $\alpha_{N,s}$ for a fixed pair (N, s) consists of two steps:

1. Generate uniformly distributed tuples $X_j = (x_1, \ldots, x_N) \in \mathcal{D}_N$ with lengths at most 1 (cf. Corollary 5.2) and compute the corresponding values $A_j := A_{N,s}(X_j)$. The $\min_j A_j$ is taken as an initial approximation of $\alpha_{N,s}$.

2. Then, each approximation of the minimizer is further optimized using the BFGS algorithm (its implementation in the `NumPy` framework has been used).

These steps are repeated iteratively to increase the likelihood of finding a global minimizer rather than becoming trapped in a local one.

Below are the results for different values (N, s)

(N, s)	Viewpoint 1	Viewpoint 2	$\approx \alpha_{N,s}$
$(3, 2)$			$0.5773867076153593 \approx 1/\sqrt{3}$
$(4, 2)$			$0.6139125728227193 \approx \sqrt{6}/4$
$(5, 2)$			0.6508046065455614
$(6, 2)$			0.6724935391390234

The conjecture that $\alpha_{3,2} = 1/\sqrt{3}$ has actually been proven (Theorem 5.13).

One also observes that, at least for small N, the points lie on the unit sphere. The following numerical simulations show that this should not be the case for $N = 5, 7, 9, 10, 11$ and for $N \geq 13$ independent from s. It is evident from the following plots where the x-axis represents N and each dot corresponds to a radius $|x_k|$ of a point of the approximate minimizer of $\alpha_{N,s}$. The red lines represent the average $\frac{1}{N}\sum_{k=1}^{N} |x_k|$ radius.

Figure 5.1: s = 2

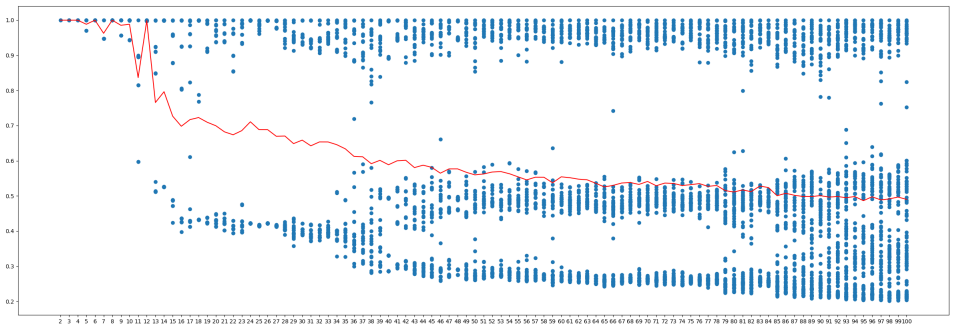


Figure 5.2: s = 3

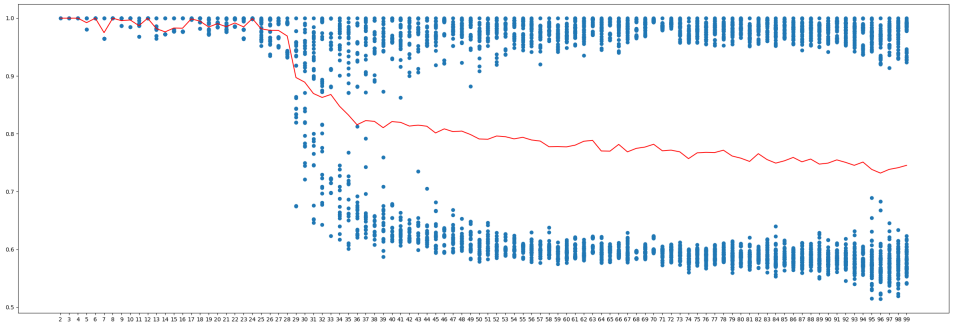
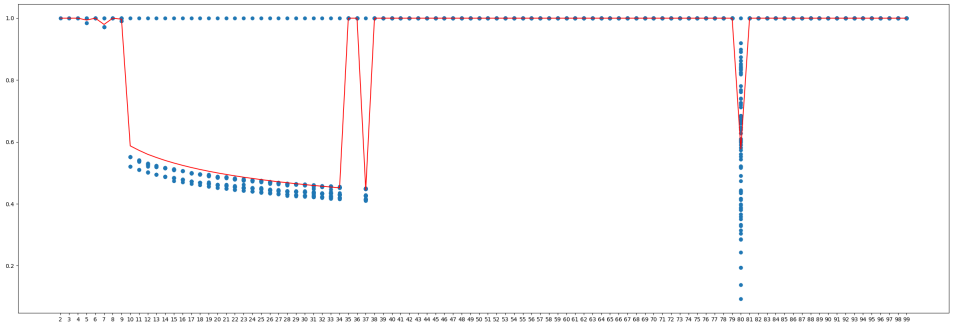


Figure 5.3: s = 10



Although there definitely are numerical errors (especially in the case $s = 10$), one observes that there are multiple radii in minimizers for $N = 5, 7$ independent of $s > 1$. It is actually possible to provide a sufficient condition for this and, by applying it, to prove the $N = 5$ case. This is achieved in the next section.

5.3 Connection to Thomson's Problem

Following numerical experiments and observations of the lengths of points of a minimizer, a natural question arises whether the minimizers of $\alpha_{N,s}$ are related to the classical Thomson's problem:

$$\mathbb{T}_N := \inf_{x_1,\dots,x_N \in \mathcal{S}^2} \sum_{1\le k<l\le N} \frac{1}{|x_k - x_l|}.$$

Note that if

$$|x_1| = |x_2| = \cdots = |x_N|, \tag{5.1}$$

then $A_{N,s}(X)$ simplifies to

$$\binom{N}{2} A_{N,s}(X) = \sum_{1\le k<l\le N} \frac{1}{|x_k - x_l|}.$$

This proves

Proposition 5.3. $\binom{N}{2}\alpha_{N,s} \le \mathbb{T}_N$ *for* $N \ge 2$, $s \ge 1$.

Remark 5.4. Note that this bound is independent from the choice of the weight w (here: $w(x) = |x|^s$) as long it is sufficiently regular, positive, and radial. By that we mean carrying out the same calculations as in Section 4.1 with some $w(x_k)$ instead of $|x_k|^s$ for a sufficiently regular $w : \mathbb{R}^3 \to (0,\infty)$. This would yield

$$\alpha_{N,w}(N-1) < Z - \frac{\operatorname{Re}\langle w(x_1)\psi, P_1^2\psi\rangle}{\langle w(x_1)|x_1|^{-1}\psi,\psi\rangle}, \tag{5.2}$$

where

$$\alpha_{N,w} := \inf\left\{ \frac{\displaystyle\sum_{1\le j<k\le N} \frac{w(x_j)+w(x_k)}{|x_j-x_k|}}{(N-1)\displaystyle\sum_{k=1}^{N}\frac{w(x_k)}{|x_k|}} \;:\; x_1,\dots,x_N \in \mathbb{R}^3 \right\}.$$

If w is radial, then restricting to $|x_k| = 1$ yields an upper bound

$$\alpha_{N,w} \le \frac{2w(1)\,\mathbb{T}_N}{(N-1)Nw(1)} = \frac{2\mathbb{T}_N}{N(N-1)} < 1.$$

Since the bound we could obtain from this equation is of the form $N < (\alpha_{N,w})^{-1}Z + o(Z)$, this means that, for radial w, the coefficient of Z will always be strictly larger than 1. It is still an open question if some non-radial weight could improve existing results with $w(x) = |x|^s$ and/or yield 1 as the coefficient.

It is then clear that, for some $N \ge 2$, the existence of $X = (x_1,\dots,x_N) \in \mathcal{D}_N$ satisfying (5.1) and $\alpha_{N,s} = A_{N,s}(X)$ is *equivalent* to $\binom{N}{2}\alpha_{N,s} = \mathbb{T}_N$.

Numerical approximations above suggest (and Theorems 5.12, 5.13 confirm) that there are indeed cases when (5.1) holds for an $\alpha_{N,s}$ minimizer. But numerical approximations also suggest that this can*not* be the case in general. In particular, Figures 5.1, 5.2, and 5.3 indicate that not all the points of a minimizer of $\alpha_{5,s}$ should lie on the unit sphere. The rest of this section is dedicated to providing a sufficient condition (Proposition 5.8) for minimizers not to satisfy (5.1) and to applying it to the case $N = 5$ (Theorem 5.10).

Let $F, G : \mathcal{D}_N \to (0, \infty)$ defined by

$$F_s(x_1, x_2, \ldots, x_N) := \sum_{1 \le k < l \le N} \frac{|x_k|^s + |x_l|^s}{|x_k - x_l|},$$
$$G_s(x_1, x_2, \ldots, x_N) := \sum_{j=1}^{N} |x_j|^{s-1}.$$

Then

$$\alpha_{N,s} = \frac{1}{N-1} \cdot \inf_X \frac{F_s(X)}{G_s(X)}.$$

Convention 5.5. From now on and until the end of this subsection, fix both $N \ge 2$ and $X = (x_1, x_2, \ldots, x_N) \in \mathcal{D}_N$ which would be a minimizer of the classical Thomson's problem. Also fix $\delta > 0$ sufficiently small such that $(1 - \varepsilon)x_1 \ne x_j$ for $|\varepsilon| < \delta$ and all $j = 2, ..., N$.

Definition 5.6. We perturb the first point by some small $|\varepsilon| < \delta$ which is represented by the following maps:

- Let $\mathcal{F}_s, \mathcal{G}_s : (-\delta, \delta) \to \mathbb{R}$,

$$\mathcal{F}_s(\varepsilon) := F_s((1 - \varepsilon)x_1, x_2, \ldots, x_N),$$
$$\mathcal{G}_s(\varepsilon) := G_s((1 - \varepsilon)x_1, x_2, \ldots, x_N).$$

- Let $a_k := \sum_{j \ne k} \frac{1}{|x_j - x_k|}$. Then we have $\mathbb{T}_N = \frac{1}{2} \sum_{k=1}^{N} a_k$.

Remark 5.7. To provide some intuition, we note that

- $\mathcal{F}_s(0) = 2\mathbb{T}_N = \sum_{k=1}^{N} a_k$.
- $\frac{2}{N}\mathbb{T}_N$ is the mean value of a_k.

Proposition 5.8 (Sufficient Condition)**.** *Let $s > 1$. The following are equivalent:*

1. $\partial_\varepsilon(\mathcal{F}_s/\mathcal{G}_s)(0) < 0$.
2. *There exist j, k such that $a_k \ne a_j$.*

Thus, the sufficient condition for an $\alpha_{N,s}$-minimizer to not be a Thomson-minimizer is that $a_k \ne a_j$ holds for some k, j.

Proof. Elementary calculations (which can be found in Appendix A.4) show that

$$\partial_\varepsilon \mathcal{F}_s \cdot \mathcal{G}_s - \mathcal{F}_s \cdot \partial_\varepsilon \mathcal{G}_s \Big|_{\varepsilon=0} = \big(-N a_1 + \sum_{k=1}^{N} a_k\big)(s - 1),$$

which is strictly less than zero if and only if $a_1 > \frac{1}{N} \sum_{k=1}^{N} a_k$. This happens if and only if there exist j, k such that $a_k \ne a_j$ (since we can relabel the points so that $j = 1$). □

Numerical simulations (based on [12]) and heuristics suggest that the condition $a_j \ne a_k$ for some j, k is indeed satisfied for Thomson-minimizers for $N \ge 13$ and $N \ne 24$, although it is still an open problem:

Conjecture 5.9. The minimizers $\hat{x}_1, ..., \hat{x}_N \in S^2$ of the classical Thomson's problem for $N \geq 13, N \neq 24$ satisfy $\sum_{i\neq k} \frac{1}{|\hat{x}_i - \hat{x}_k|} \neq \sum_{i\neq j} \frac{1}{|\hat{x}_i - \hat{x}_j|}$ for at least one pair of indices j, k.

Since the unique Thomson minimizer for $N = 5$ is proven [10] to be the triangular bi-pyramid, one easily checks that the condition $a_j \neq a_k$ for some j, k is satisfied. This shows

Theorem 5.10. *The minimizers of $\mathbb{T}_5$ and $\alpha_{5,s}$ do not coincide for all $s > 1$, thus, if $\{\hat{x}_j\}$ minimize $\alpha_{5,s}$, then $\#\{|\hat{x}_j| : 1 \leq j \leq N\} > 1$. Moreover, $\alpha_{5,s} < \frac{1}{10}\mathbb{T}_5$.*

This theorem explains the observations made during numerical experiments.

5.4 Universal Lower Bound

Numerical experiments also suggest that $\alpha_{N,s} \geq 1/2$ for all $N \in \mathbb{N}_{\geq 2}, s \geq 1$. We prove it in this subsection. This fact connects the general case $s > 1$ to the method used by E. Lieb which proves $N < 2Z + 1$ since the generalized approach yields a bound of the form $N < (\alpha_{N,s})^{-1}Z +$ (lower order terms).

Proposition 5.11. $\alpha_{N,s} \geq 1/2$ *for* $N \geq 2, s \geq 1$.

Proof. First, observe that

$$\frac{a^s + b^s}{a + b} \geq \frac{a^{s-1} + b^{s-1}}{2} \qquad a, b > 0. \tag{5.3}$$

Indeed,

$$\begin{aligned} (5.3) &\iff a^s + b^s \geq ba^{s-1} + ab^{s-1} \\ &\iff (a - b)(a^{s-1} - b^{s-1}) \geq 0. \end{aligned}$$

By definition,

$$\alpha_{N,s} = \inf_{x_1,...,x_N \in \mathbb{R}^3} \frac{1}{(N-1)\sum_{j=1}^N |x_j|^{s-1}} \sum_{1\leq k<l\leq N} \frac{|x_k|^s + |x_l|^s}{|x_k - x_l|}.$$

Then, using the triangle inequality and (5.3), one has

$$\frac{|x_k|^s + |x_l|^s}{|x_k - x_l|} \geq \frac{|x_k|^s + |x_l|^s}{|x_k| + |x_l|} \geq \frac{1}{2}(|x_k|^{s-1} + |x_l|^{s-1})$$

for $1 \leq k < l \leq N$. Thus

$$\sum_{1\leq k<l\leq N} \frac{|x_k|^s + |x_l|^s}{|x_k - x_l|} \geq \frac{1}{2} \sum_{1\leq k<l\leq N} (|x_k|^{s-1} + |x_l|^{s-1}) = \frac{1}{2}(N-1)\sum_{j=1}^N |x_j|^{s-1}$$

for any configuration of pairwise distinct points $x_1, \ldots, x_N \in \mathbb{R}^3$. This implies the claim. □

5.5 Concrete Values

Theorem 5.12. *$\alpha_{2,s} = 1/2$ for all $s \geq 1$.*

Proof. We need to find the minimum of

$$\frac{|x|^s + |y|^s}{|x-y|(|x|^{s-1} + |y|^{s-1})}.$$

According to Corollary 5.2, we may assume that $x = e_1$. We set $a := |y|$. Then by the law of cosines $|x-y| = \sqrt{a^2 + 1 - 2a\cos\theta}$ for some $\theta \in [0, \pi]$. Now the following expression is to be minimized:

$$\frac{a^s + 1}{\sqrt{a^2 + 1 - 2a\cos\theta}\,(a^{s-1} + 1)}, \quad a > 0.$$

Obviously, we need $\theta = \pi$, and thus $-2a\cos\theta = 2a$, i.e. the points are antipodal. Thus

$$\alpha_{2,s} = \inf_{a>0} \frac{a^s + 1}{(a+1)(a^{s-1} + 1)}, \quad \text{since } a + 1 > 0.$$

It is an easy calculation to show that the derivative of $s \mapsto \frac{a^s+1}{(a+1)(a^{s-1}+1)}$ is positive for all fixed $a \neq 1$ and is zero for $a = 1$. Thus, $\alpha_{2,s} = \alpha_{2,1} = 1/2$. □

Theorem 5.13. *$\alpha_{3,2} = 1/\sqrt{3}$, where the minimum is attained when the three points form an equilateral triangle inscribed in the unit circle.*

Proof. Upper bound: $\alpha_{3,2} \leq 1/\sqrt{3}$ since $\alpha_{N,s}$ by definition is not greater than $\frac{1}{(N-1)\sum_{j=1}^N |x_j|^{s-1}} \sum_{1\leq k<l\leq N} \frac{|x_k|^s + |x_l|^s}{|x_k - x_l|}$ for any $(x_1, \ldots, x_N) \in \mathcal{D}_N$, $N \geq 2$, $s \geq 1$.

A direct calculation with $x_j = (\cos\frac{2\pi}{3}j, \sin\frac{2\pi}{3}j, 0)$ shows $\alpha_{3,2} \leq \frac{1}{2} \cdot \frac{3\cdot 2}{\sqrt{3}\cdot 3} = 1/\sqrt{3}$.

Lower bound: Let $x_a, x_b, x_c \in \mathbb{R}^3$, $x_j \neq x_k$ for $j \neq k$ be a minimizer of $\alpha_{3,2}$ (here: a, b, c are used as formal indices). We also write $a = |x_1|$, $b = |x_2|$, $c = |x_3|$ and $d_{ab} = |x_a - x_b|$ and similarly d_{bc}, d_{ac} (it is clear from the context whether a letter a, b, or c represents an index or the corresponding length). Since the problem is scale invariant (cf. Lemma 5.1), we are free to assume $a + b + c = 1$. Then we have, for $N = 3$, $s = 2$:

$$2\alpha_{3,2} = \frac{a^2 + b^2}{d_{ab}} + \frac{b^2 + c^2}{d_{bc}} + \frac{a^2 + c^2}{d_{ac}}.$$

By a special case of CSI (in this case referred to as Engel's inequality, cf. Appendix A.1),

$$\left(\sqrt{a^2 + b^2} + \sqrt{b^2 + c^2} + \sqrt{a^2 + c^2}\right)^2 \leq 2\alpha_{3,2}(d_{ab} + d_{bc} + d_{ac}).$$

Equivalently,

$$2\alpha_{3,2} \geq \frac{\left(\sqrt{a^2 + b^2} + \sqrt{b^2 + c^2} + \sqrt{a^2 + c^2}\right)^2}{d_{ab} + d_{bc} + d_{ac}}.$$

We also have

$$d_{ab} + d_{bc} + d_{ac} \leq 3\sqrt{a^2 + b^2 + c^2}, \tag{5.4}$$

$$\left(\sqrt{a^2 + b^2} + \sqrt{b^2 + c^2} + \sqrt{a^2 + c^2}\right)^2 \geq 3(a^2 + b^2 + c^2) + 1. \tag{5.5}$$

The proofs of (5.4) and (5.5) can be found in Appendix A.2. If we denote $S_2 = a^2 + b^2 + c^2$, then

$$2\alpha_{3,2} \geq \frac{3S_2 + 1}{3\sqrt{S_2}} = f(\sqrt{S_2}) \geq \frac{2}{\sqrt{3}}$$

for $f(x) = x + \frac{1}{3x}$ with $\min_{x>0} f(x) = f(1/\sqrt{3}) = 2/\sqrt{3}$. □

5.6 Asymptotic Behavior

Theorem 5.14. *For all $N \in \mathbb{N} \setminus \{1\}$ we have $\lim_{s\to\infty} \alpha_{N,s} = 1/2$.*

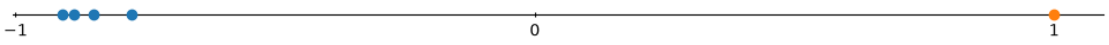
Proof. We have $\alpha_{2,s} = 1/2$ for all $s > 1$ due to Theorem 5.12. Hence, fix some $N \geq 3$.

Claim: for each $s > 1$, there exists a configuration of points $X_s = (x_1^{(s)}, x_2^{(s)}, \ldots, x_N^{(s)})$ such that $A_{N,s}(X_s) \to 1/2$ for $s \to \infty$.

We proceed as follows:

- $x_1^{(s)} := e_1 = (1, 0, 0)^t$
- $x_j^{(s)} := -e_1 + \delta_j^{(s)} e_1 = e_1(\delta_j^{(s)} - 1)$ for $j \geq 2$, where $\delta_j^{(s)}$ will be chosen such that the points $x_j^{(s)}$ $(j \geq 2)$ accumulate near $-e_1$ (and converge towards it, i.e. $\lim_{s\to\infty} \delta_j^{(s)} = 0$) but are all distinct.

The following figure demonstrates this construction for $N = 5, s = 5$:



A naive way is to choose $\delta_j^{(s)} = 1/(sj)$, but this will not work due to convergence issues. For now, we assume that we have a strictly increasing divergent "sequence" $\{m_s\}_{s>1}$ that we will choose later, and set $\delta_j^{(s)} := \frac{1}{m_s} \cdot \frac{1}{j}$.

Now, observe that

- $|x_j^{(s)}| = 1 - \delta_j^{(s)} = 1 - \frac{1}{m_s \cdot j}$, $j \geq 2$
- $|x_k^{(s)} - x_1^{(s)}| = 2 - \delta_k^{(s)}$, $k \geq 2$
- $|x_k^{(s)} - x_l^{(s)}| = |\delta_k^{(s)} - \delta_l^{(s)}| = \left(\frac{1}{k} - \frac{1}{l}\right) \frac{1}{m_s}$, where $2 \leq k < l \leq N$.

To obtain a lower bound of the last expression, note that

$$\min_{2\leq k<l\leq N} \left(\frac{1}{k} - \frac{1}{l}\right) = \min_{2<l\leq N} \left(\frac{1}{l-1} - \frac{1}{l}\right) = \frac{1}{N(N-1)},$$

hence

$$|x_k^{(s)} - x_l^{(s)}| \geq \frac{1}{N(N-1)} \cdot \frac{1}{m_s}. \tag{1}$$

We write

$$\phi_{k,l}^{(s)} := \frac{|x_k^{(s)}|^s + |x_l^{(s)}|^s}{\sum_{j=1}^N |x_j^{(s)}|^{s-1}} \qquad (1 \leq k, l \leq N),$$

then

$$\begin{aligned}
(N-1)A_{N,s}(X_s) &= \sum_{1\leq k<l\leq N} \phi_{k,l}^{(s)} \cdot \frac{1}{|x_k^{(s)} - x_l^{(s)}|} \\
&= \underbrace{\sum_{k=2}^{N} \phi_{k,1}^{(s)} \cdot \frac{1}{|x_1^{(s)} - x_k^{(s)}|}}_{D_s} + \underbrace{\sum_{2\leq k<l\leq N} \phi_{k,l}^{(s)} \cdot \frac{1}{|x_k^{(s)} - x_l^{(s)}|}}_{E_s} \\
&= D_s + E_s.
\end{aligned}$$

Using the estimates above, we get

$$\begin{aligned} E_s &= \sum_{2\le k<l\le N} \phi_{k,l}^{(s)} \cdot \frac{1}{|x_k^{(s)} - x_l^{(s)}|} \\ &\le \sum_{2\le k<l\le N} \phi_{k,l}^{(s)}\, m_s N(N-1). \end{aligned}$$

Observe that, by construction, $|x_k^{(s)}| < |x_l^{(s)}|$ for $2 \le k < l$. Thus, an upper bound for $\phi_{k,l}^{(s)}$ (where $k < l$) is obtained the following way:

$$\begin{aligned} \phi_{k,l}^{(s)} &= \frac{|x_k^{(s)}|^s + |x_l^{(s)}|^s}{\sum_{j=1}^N |x_j^{(s)}|^{s-1}} \\ &= \frac{|x_k^{(s)}|^s + |x_l^{(s)}|^s}{1 + \sum_{j=2}^N |x_j^{(s)}|^{s-1}} \le \frac{|x_k^{(s)}|^s + |x_l^{(s)}|^s}{1 + |x_k^{(s)}|^{s-1} + |x_l^{(s)}|^{s-1}} \le 2|x_l^{(s)}|^s. \end{aligned}$$

Thus,

$$\begin{aligned} E_s &\le \sum_{2\le k<l\le N} 2|x_l^{(s)}|^s\, m_s N(N-1) \\ &\le \sum_{2\le k<l\le N} 2|x_N^{(s)}|^s\, m_s N(N-1) \\ &= \underbrace{2N(N-1)\frac{(N-1)(N-2)}{2}}_{=:C} |x_N^{(s)}|^s m_s = C|x_N^{(s)}|^s m_s. \end{aligned}$$

Similarly,

$$D_s = \sum_{k=2}^N \frac{1 + |x_k^{(s)}|^s}{1 + \sum_{j=2}^N |x_k^{(s)}|^{s-1}} \cdot \underbrace{\frac{1}{2 - \delta_k^{(s)}}}_{\substack{\to 1/2 \text{ as } s\to\infty \\ \text{by construction}}}.$$

So: in order to have $A_{N,s}(X_s) \to 1/2$, we need $E_s \to 0$ and $D_s \to (N-1)/2$ as $s \to \infty$. This would be achieved if $|x_j^{(s)}|^s m_s \to 0$ for all $j \ge 2$ (and thus $|x_j^{(s)}|^s \to 0$, as well as $|x_j^{(s)}|^{s-1} \to 0$).

Remember that $|x_j^{(s)}| = 1 - \frac{1}{m_s j}$, and m_s must be strictly increasing in s with $m_s \to \infty$. Consider

$$\begin{aligned} |x_j^{(s)}|^s = \Big(1 - \frac{1}{m_s j}\Big)^s &= \Big[\Big(1 - \frac{1}{m_s j}\Big)^{m_s j}\Big]^{s/(m_s j)} \\ &\le (e^{-s/m_s})^{1/j} \end{aligned}$$

Thus, choosing $m_s = \sqrt{s}$, implies $|x_j^{(s)}|^s m_s \to 0$ as $s \to \infty$ for all $j \ge 2$. Now, we combine both estimates and obtain

$$(N-1)A_{N,s}(X_s) = D_s + E_s \le C|x_N^{(s)}|^s m_s + D_s \to 0 + \frac{N-1}{2} \qquad (s \to \infty),$$

hence

$$\frac{1}{2} \le \alpha_{N,s} \le A_{N,s} \to \frac{1}{2} \qquad (s \to \infty),$$

where the lower bound $1/2 \le \alpha_{N,s}$ has been proven in Proposition 5.11. □

This theorem outlines geometric properties of the minimizers of $\alpha_{N,s}$ for large s and suggests that, for each N, the supremum $\sup\{\alpha_{N,s} : s > 1\}$ should be attained at a finite value of s. Consequently, to obtain the tightest bound on the maximal excess charge via (4.1), one indeed has to consider some finite $s > 1$.

Note that this observation is consistent with [5] where only $1 < s \leq 3$ was considered, (although due to another technical limitation). This will be discussed in the next chapter.

Chapter 6

Mean-Field Type Bounds and Refinements

Previously, we derived—cf. (4.8)—that

$$\alpha_{N,s}(N-1) < Z + \frac{s^2-1}{8}\, C_{s-1}^{-1/(s-1)} \kappa Z N^{-2/3}$$

is a necessary condition for binding, and the goal is now to obtain the tightest lower bound on the left-hand side for the most optimal choice of $s \geq 2$. The previous section proved a universal lower bound $\frac{1}{2} \leq \alpha_{N,s}$ which recovers E. Lieb's result with the coefficient 2 before Z, but it is possible to refine the method using mean-field type bounds. This is the objective of this chapter.

Recall the definition of $\alpha_{N,s}$:

$$\alpha_{N,s} := \inf \left\{ \frac{\displaystyle\sum_{1\leq j<k\leq N} \frac{|x_j|^s + |x_k|^s}{|x_j - x_k|}}{(N-1)\displaystyle\sum_{k=1}^{N} |x_k|^{s-1}} \;:\; x_1, \ldots, x_N \in \mathcal{D}_N \right\}.$$

We use its mean-field type approximation introduced in [5] (based on [8], originally only for the case $s = 2$):

$$\beta_s := \inf \left\{ \frac{\displaystyle\iint \frac{|x|^s + |y|^s}{2|x-y|}\, \mathrm{d}\rho(x)\, \mathrm{d}\rho(y)}{\displaystyle\int |x|^{s-1}\, \mathrm{d}\rho(x)} \;:\; \rho \in D(\beta_s) \right\},$$

where $D(\beta_s) := P(\mathbb{R}^3) \cap H^{-1}(\mathbb{R}^3) \cap L_{s-1}(\mathbb{R}^3)$. Here, $P(\mathbb{R}^3)$ is the set of all probability measures on $\mathbb{R}^3$, $H^{-1}(\mathbb{R}^3)$ is the dual of the usual Sobolev space $H^1(\mathbb{R}^3)$, and $L_t(\mathbb{R}^3)$ is the set of all finite signed or complex–valued measures on $\mathbb{R}^3$ for which $\int |x|^t \,\mathrm{d}|\mu|(x) < \infty$.

We start by showing the first important property (due to [5] based on [8]) connecting the two quantities.

Lemma 6.1. *For all $N \in \mathbb{N}_{\geq 2}$ and $s \geq 1$, one has $\alpha_{N,s} \leq \beta_s$.*

Proof. By symmetry,

$$\iint_{\mathbb{R}^3\times\mathbb{R}^3} \frac{|x|^s+|y|^s}{2|x-y|}\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(y) = \frac{1}{N}\int_{\mathbb{R}^{3N}} \frac{1}{N-1}\sum_{1\le j<k\le N} \frac{|x_j|^s+|x_k|^s}{|x_j-x_k|}\,\mathrm{d}\mu(x_1)\cdots\mathrm{d}\mu(x_N)$$
$$\ge \frac{\alpha_{N,s}}{N}\int_{\mathbb{R}^{3N}}\sum_{k=1}^N |x_k|^{s-1}\,\mathrm{d}\mu(x_1)\cdots\mathrm{d}\mu(x_N) = \alpha_{N,s}\int_{\mathbb{R}^3}|x|^{s-1}\,\mathrm{d}\mu(x).$$

Thus $\alpha_{N,s}\le\beta_s$ for all $N\in\mathbb{N}_{\ge 2}$ and $s\ge 1$. □

Another important observation is the following lemma. Its version for $s=2$ had initially been proven in [8]. It was extended to all $s\ge 1$ in [5].

Lemma 6.2. *$\alpha_{N,s}$ is increasing in N for all $s\ge 1$.*

Proof. Singling out the m-th particle, we have

$$\sum_{1\le j<k\le N}\frac{|x_j|^s+|x_k|^s}{|x_j-x_k|} = \sum_{m=1}^N\left(\frac{1}{N-2}\sum_{\substack{1\le j<k\le N\\ j\neq m,k\neq m}}\frac{|x_j|^s+|x_k|^s}{|x_j-x_k|}\right)$$
$$\ge \sum_{m=1}^N\left(\alpha_{N-1,s}\sum_{\substack{k=1\\k\neq m}}^N|x_k|^{s-1}\right) = \alpha_{N-1,s}(N-1)\sum_{k=1}^N|x_k|^{s-1}.$$

The claim follows from the definition of $\alpha_{N,s}$. □

With these two lemmas, it is natural to conjecture that $\alpha_{N,s}$ might converge to β_s as $N\to\infty$. This turns out to be the case and is proved using the following result from [5].

Lemma 6.3. *For every $N\ge 2, r\in(0,1)$, and $s\ge 0$,*

$$\frac{(1+r)^{s+1}-(1-r)^{s+1}}{2r(s+1)}N\beta_s \le (1+r)^s\left(\alpha_{N,s}(N-1)+\frac{1}{r}\right). \tag{6.1}$$

The proof is provided in Appendix A.6. As an immediate consequence, one has

Corollary 6.4. *$\alpha_{N,s}\to\beta_s$ as $N\to\infty$ for $s>0$.*

Proof. Lemma 6.3 shows

$$\liminf_{N\to\infty}\alpha_{N,s}\ge\frac{(1+r)^{s+1}-(1-r)^{s+1}}{2r(s+1)(1+r)^s}\beta_s.$$

Taking the limit $r\to 0$ yields $\liminf_{N\to\infty}\alpha_{N,s}\ge\beta_s$ and, together with Lemma 6.1, this proves

$$\lim_{N\to\infty}\alpha_{N,s}=\beta_s,\quad s>0$$

for all $s>0$. □

Although Lemma 6.3 is conceptually important, it is too crude for a good lower bound on $\alpha_{N,s}$ since it ignores the angular dependence. To obtain a sharper lower bound on $\alpha_{N,s}$, significantly more refined analysis is required. This was done by P.T. Nam in [8] for $s=2$. In [5], two successive refinements are developed for general $s>1$:

1. **Refined Comparison I (Lemma 5.5 in [5])**: Uses a convexity argument to bound the angular numerator $(1+qt)^{s/2}$ by a quadratic polynomial. This improves the prefactor of $\alpha_{N,s}$ but still relies on a relatively crude estimate of the resulting integrals.

2. **Refined Comparison II (Lemma 5.9 in [5])**: Employs a full *multipole expansion* of the Coulomb kernel in terms of Legendre polynomials. This resolves the angular integrals exactly, yielding the sharpest possible bound for the range $2 \leq s \leq 3$.

The second refinement supersedes the first and is the one that is applied in the final minimization. We therefore state it as the central tool for the final bound.

Lemma 6.5 (Sharp comparison of $\alpha_{N,s}$ and β_s, [5])**.** *Let $2 \leq s \leq 3$. For every $N \geq 2$ and $r > 0$, we have*

$$\frac{(1+r)^{s+1}-(1-r)^{s+1}}{2r(s+1)} N\beta_s \leq \frac{(1+r)^{s+2}-(1-r)^{s+2}}{2r(s+2)} \left(\alpha_{N,s}(N-1)+\frac{1}{r}\right) + r^2 f(r,s)\alpha_{N,s}(N-1), \tag{6.2}$$

where

$$f(r,s) := \frac{s}{2}\left(\frac{s}{2}-1\right)\left(\frac{4}{15}+\left(2-\frac{s}{2}\right)\frac{8}{105}r+\left(2-\frac{s}{2}\right)\frac{448}{625}r^2\right).$$

The next step would be to obtain a lower bound on β_s to be used in (6.2). This is done in the next section by utilizing the radial symmetry of minimizing sequences of β_s for $1 \leq s \leq 3$. We will also show that that argument breaks for $s > 3$ which is an original result.

But for now, we assume that we have a lower bound $b^{-1}(s) \leq \beta_s$. The goal is to solve (6.2) for N to obtain the following upper bound on the maximum excess charge of electrons which is the main result (Theorem 2.2) of [5]:

Theorem 6.6. *Let $s \in (1,3]$. Then there exists $c(s) > 0$ such that, if $E_{N,Z} < E_{N-1,Z}$, then*

$$N < b(s)\, Z + c(s)\, Z^{1/3}.$$

Remark 6.7. It will also be shown in the next section that $b(s) = \max\limits_{0\leq t\leq 1} \dfrac{1+t^{s-1}}{1+t^s}$ can be chosen for $s \in (1,3]$. Note that it converges to 1 as $s \to \infty$, whereas $c(s)$ diverges.

Proof. Note that the fraction on the left-hand side of (6.2) is positive for all $r > 0$ and $s > 0$ and, consequently, we can use the lower bound $b(s)^{-1}$ on β_s. Since $(1+r)^p-(1-r)^p = \sum_{k=1}^{N} \binom{p}{k} 2\chi(k \text{ is odd}) r^k = 2rp + (\text{non-negative terms}) \geq 2rp$ for $r \in [0,1], p \geq 2$. Hence

$$\frac{2r(s+2)}{(1+r)^{s+2}-(1-r)^{s+2}} \leq 1.$$

Together with $f(r,s) < 1/2$ [5] this yields

$$\frac{2r(s+2)}{(1+r)^{s+2}-(1-r)^{s+2}} r^2 f(r,s) \leq \frac{r^2}{2}.$$

Multiplying both sides of (6.2) by $\frac{2r(s+2)}{(1+r)^{s+2}-(1-r)^{s+2}}$, we conclude

$$\frac{s+2}{s+1}\frac{(1+r)^{s+1}-(1-r)^{s+1}}{(1+r)^{s+2}-(1-r)^{s+2}}\frac{N}{b(s)}-\frac{1}{r}\leq\alpha_{N,s}(N-1)\left(1+\frac{r^2}{2}\right).$$

A technical result from [5] (Lemma A.8) shows that

$$\frac{s+2}{s+1}\frac{(1+r)^{s+1}-(1-r)^{s+1}}{(1+r)^{s+2}-(1-r)^{s+2}}\geq 1-\frac{s}{3}r^2.$$

Thus we obtain

$$\left(1-\frac{s}{3}r^2\right)\frac{N}{b(s)}-\frac{1}{r}<\alpha_{N,s}(N-1)\left(1+\frac{r^2}{2}\right).$$

To maximize the lower bound, we choose $r=\left(\frac{3}{2s}\right)^{1/3}\left(\frac{N}{b(s)}\right)^{-1/3}$ and conclude

$$\frac{N}{b(s)}\leq\alpha_{N,s}(N-1)\left(1+\frac{\lambda^2}{2}N^{-2/3}\right)+\left(\lambda^{-1}+\frac{s}{3}\frac{\lambda^2}{b(s)}\right)N^{1/3}.$$

Applying Lemma 4.4 shows

$$\begin{aligned}\frac{N}{b(s)}&\leq Z\left(1+AN^{-2/3}\right)\left(1+\frac{\lambda^2}{2}N^{-2/3}\right)+\left(\lambda^{-1}+\frac{s}{3}\frac{\lambda^2}{b(s)}\right)N^{1/3}=\\&=Z+\left(A+\frac{\lambda^2}{2}\right)ZN^{-2/3}+\left(\lambda^{-1}+\frac{s}{3}\frac{\lambda^2}{b(s)}\right)N^{1/3}+\frac{\lambda^2 A}{2}ZN^{-4/3},\end{aligned}$$

where

$$A:=\frac{s^2-1}{8}C_{s-1}^{-1/(s-1)}\kappa$$

is the parameter in Lemma 4.4. By E. Lieb's bound, $Z\leq N\leq 3Z$, and thus, if $E_{N,Z}<E_{N-1,Z}$, then there exists some $c(s)>0$ such that

$$\boxed{N<b(s)Z+c(s)Z^{1/3}}$$

□

Then, cases $s=2,3$ can be covered through finer calculations to obtain improved bounds on $c(s)$.

Proposition 6.8 (Prop. 2.4 in [5]). *For any $Z\geq 2$, if $E_{N,Z}<E_{N-1,Z}$, then*

$$\boxed{N<\frac{1}{2}(\sqrt{2}+1)\,Z+2.96\,Z^{1/3}}$$

where

$$1.2071<\frac{1}{2}(\sqrt{2}+1)<1.2072.$$

Proposition 6.9 (Prop. 2.5 in [5]). *For any $Z\geq 4$, if $E_{N,Z}<E_{N-1,Z}$, then*

$$\boxed{N<b(3)\,Z+3.90\,Z^{1/3}+0.0134+0.184\,Z^{-1/3}+0.0196\,Z^{-2/3}}$$

where

$$1.1184<b(3)=\frac{2}{3}\frac{\sqrt[3]{1+\sqrt{2}}}{(1+\sqrt{2})^{2/3}-1}<1.1185.$$

The next step is to find a suitable $b(s)$.

6.1 Symmetry Arguments

As a tool for analyzing β_s, we define the bilinear map

$$F_s(\mu,\nu) := \iint_{\mathbb{R}^3\times\mathbb{R}^3} \frac{|x|^s+|y|^s}{2|x-y|}\,\mathrm{d}\mu(x)\,\mathrm{d}\nu(y), \qquad \mu,\nu \text{ measures on } \mathbb{R}^3.$$

The rest of this chapter is dedicated to its investigation to obtain a lower bound on β_s.

Assume that μ is rotationally symmetric. Then we can apply Tonelli's and Newton's theorems:

$$\begin{aligned} F_s(\mu,\nu) &= \int_{\mathbb{R}^3}\left(\int_{\mathbb{R}^3}\frac{|x|^s+|y|^s}{2|x-y|}\,\mathrm{d}\mu(x)\right)\mathrm{d}\nu(y) \\ &= \int_{\mathbb{R}^3}\left(\int_{\mathbb{R}^3}\frac{|x|^s+|y|^s}{2\max(|x|,|y|)}\,\mathrm{d}\mu(x)\right)\mathrm{d}\nu(y). \end{aligned}$$

Now, take an arbitrary $\mu\in D(\beta_s)$. We define μ_0 to be the rotational average of μ:

$$\int_{\mathbb{R}^3} f(x)\,\mathrm{d}\mu_0(x) := \frac{1}{|S^2|}\int_{\mathbb{R}^3}\int_{S^2} f(|x|w)\,d\omega\,\mathrm{d}\mu(x).$$

In particular, it does not change the value of the integral $\int \mathrm{d}\mu = \int \mathrm{d}\mu_0$ since the Lebesgue measure is rotationally symmetric. So, using the bilinearity of F_s, we have

$$\begin{aligned} F_s(\mu,\mu) &= F_s(\mu-\mu_0+\mu_0,\mu-\mu_0+\mu_0) \\ &= F_s(\mu-\mu_0,\mu-\mu_0)+F_s(\mu-\mu_0,\mu_0)+F_s(\mu_0,\mu-\mu_0)+F_s(\mu_0,\mu_0) \\ &= F_s(\mu-\mu_0,\mu-\mu_0)+2F_s(\mu-\mu_0,\mu_0)+F_s(\mu_0,\mu_0). \end{aligned} \tag{6.3}$$

Take some $\nu\in D(\beta_s)$. Then

$$\begin{aligned} F_s(\mu_0,\nu) &= \frac{1}{2}\iint_{\mathbb{R}^3\times\mathbb{R}^3}\frac{|x|^s+|y|^s}{\max(|x|,|y|)}\,\mathrm{d}\mu_0(x)\,\mathrm{d}\nu(y) \\ &= \frac{1}{2}\iint_{\mathbb{R}^3\times\mathbb{R}^3}\frac{|x|^s+|y|^s}{\max(|x|,|y|)}\,\mathrm{d}\mu_0(x)\,\mathrm{d}\nu_0(y). \end{aligned}$$

The last equation holds due to (6.1). Now consider $\nu=\mu-\mu_0$, i.e. we remove the angular component of μ. Then $\nu_0=\mu_0-\mu_0=0$. Thus,

$$F_s(\mu_0,\mu-\mu_0)=0.$$

Combining this with (6.3) yields

$$F_s(\mu,\mu)=F_s(\mu_0,\mu_0)+F_s(\mu-\mu_0,\mu-\mu_0).$$

Now we make a generous assumption: $F_s(\mu-\mu_0,\mu-\mu_0)\geq 0$. Then

$$\begin{aligned} \beta_s &= \inf\left\{\frac{\iint\frac{|x|^s+|y|^s}{2|x-y|}\,\mathrm{d}\rho(x)\,\mathrm{d}\rho(y).}{\int|x|^{s-1}\,\mathrm{d}\rho(x)} \;:\; \rho\in D(\beta_s)\right\} \\ &= \inf\left\{\frac{\iint\frac{|x|^s+|y|^s}{2|x-y|}\,\mathrm{d}\rho(x)\,\mathrm{d}\rho(y).}{\int|x|^{s-1}\,\mathrm{d}\rho(x)} \;:\; \rho\in D(\beta_s),\ \rho=\rho_0\right\} =: \beta_s^{\mathrm{rad}}. \end{aligned}$$

It turns out that $F_s(\mu-\mu_0,\mu-\mu_0)$ is indeed non-negative for $1\leq s\leq 3$. This result is highly significant for this section, thus we state it as a lemma:

Lemma 6.10 (Lemma 4.3 [5]). *$F_s(\mu-\mu_0,\mu-\mu_0)\geq 0$ for $1\leq s\leq 3$.*

The proof can be found in Section 6.2. Moreover, we show that, in general, this is indeed false for $s>3$ which is a new result. As a corollary, we prove the following lower bound on β_s:

Lemma 6.11 (A lower bound on the mean-field type approximation, [5]).

$$\boxed{\beta_s \geq \inf_{0<t\leq 1}\frac{1+t^s}{1+t^{s-1}} = \min_{0<t\leq 1}\frac{1+t^s}{1+t^{s-1}}, \qquad 1\leq s\leq 3.}$$

Proof. Let $\rho\in D(\beta_s)$ with $\rho=\rho_0$. Recall

$$F_s(\rho,\rho)=\frac{1}{2}\iint_{\mathbb{R}^3\times\mathbb{R}^3}\frac{|x|^s+|y|^s}{\max(|x|,|y|)}\,\mathrm{d}\rho(x)\,\mathrm{d}\rho(y).$$

Then we have for the integrand

$$\begin{aligned}\frac{|x|^s+|y|^s}{\max(|x|,|y|)} &= \frac{|x|^s+|y|^s}{\max(|x|,|y|)}\cdot\frac{|x|^{s-1}+|y|^{s-1}}{|x|^{s-1}+|y|^{s-1}}\\ &= \frac{|x|^s(1+(|y|/|x|)^s)}{|x|^s\max(1,|y|/|x|)(1+(|y|/|x|)^{s-1})}(|x|^{s-1}+|y|^{s-1})\\ &\geq \inf_{t>0}\frac{1+t^s}{\max(1,t)(1+t^{s-1})}(|x|^{s-1}+|y|^{s-1})\\ &= \underbrace{\inf_{0<t\leq 1}\frac{1+t^s}{1+t^{s-1}}}_{=:\iota(s)}(|x|^{s-1}+|y|^{s-1}).\end{aligned}$$

Hence, we can bound the bilinear form from below by

$$\begin{aligned}F_s(\rho,\rho) &\geq \frac{1}{2}\iint_{\mathbb{R}^3\times\mathbb{R}^3}\frac{|x|^s+|y|^s}{\max(|x|,|y|)}\,\mathrm{d}\rho(x)\,\mathrm{d}\rho(y)\\ &\geq \iota(s)\frac{1}{2}\iint_{\mathbb{R}^3\times\mathbb{R}^3}(|x|^{s-1}+|y|^{s-1})\,\mathrm{d}\rho(x)\,\mathrm{d}\rho(y)\\ &= \iota(s)\int_{\mathbb{R}^3}|x|^{s-1}\,\mathrm{d}\rho(x).\end{aligned}\tag{6.4}$$

Consequently,

$$\beta_s\geq\iota(s)=\inf_{0<t\leq 1}\frac{1+t^s}{1+t^{s-1}}=\min_{0<t\leq 1}\frac{1+t^s}{1+t^{s-1}},\quad 1\leq s\leq 3.$$

The function $\iota(s,t)=(1+t^s)/(1+t^{s-1})$ has derivative

$$\partial_t\,\iota(s,t)=\frac{t^{s-2}(t^s+s(t-1)+1)}{(1+t^{s-1})^2}.$$

The critical point $t_0\in(0,1)$ satisfies

$$t_0^s+st_0+1-s=0,$$

which has a unique solution in $(0,1)$ since $t\mapsto t^s+st+1-s$ is strictly increasing and changes sign. Thus

$$\beta_s\geq\iota(s,t_0)=:b(s)^{-1}.$$

□

Hence, as a necessary condition for binding, we obtain

$$\boxed{N < \iota(s)^{-1} Z + c(s) Z^{1/3}, \quad 1 \le s \le 3}$$

with $\iota(s) = \min_{0<t\le 1} \frac{1+t^s}{1+t^{s-1}}$.

6.2 Proof of Lemma 6.10

Now we prove the central property (Lemma 6.10) of the bilinear form F_s, and thus of β_s, namely that, for any $\mu \in D(\beta_s)$—and for the minimizing sequences of β_s in particular—$F_s(\mu - \mu_0, \mu - \mu_0) \ge 0$. This allowed us to only consider rotationally symmetric measures $\mu = \mu_0$. The following argument works for $1 \le s \le 3$ and absolutely continuous measures η with respect to the Lebesgue measure λ^3. The argument is different from the proof in [5] (Lemma 4.3) which does not restrict to λ^3-absolutely continuous measures, but our argument allows us to show that the positivity indeed breaks down in general for larger $s > 3$ (this is convered in the next subsection). For the proof for general $\mu \in D(\beta_s)$, refer to [5, Lemma 4.3].

Put $\eta := \mu - \mu_0$. The claim is $F_s(\eta, \eta) \ge 0$ for $s \in [1, 3]$. Recall the bilinear form

$$\begin{aligned} F_s(\eta,\eta) :&= \frac{1}{2} \iint_{\mathbb{R}^3\times\mathbb{R}^3} \frac{|x|^s + |y|^s}{|x-y|} \mathrm{d}\eta(x) \mathrm{d}\eta(y) \\ &= \iint_{|x|<|y|} \frac{|x|^s + |y|^s}{|x-y|} \mathrm{d}\eta(x) \mathrm{d}\eta(y) \end{aligned}$$

which we express in spherical coordinates as

$$F_s(\eta,\eta) = \int_0^\infty \int_{S^2} \int_{r_1}^\infty \int_{S^2} \frac{r_1^s + r_2^s}{|r_1\omega_1 - r_2\omega_2|} \eta(r_1\omega_1)\eta(r_2\omega_2)\, r_1^2\, r_2^2 \,\mathrm{d}\omega_2 \,\mathrm{d}r_2 \,\mathrm{d}\omega_1 \,\mathrm{d}r_1.$$

Now, we expand the $1/|r_1\omega_1 - r_2\omega_2|$ term with $r_1 < r_2$ using *spherical harmonics* (e.g. [1, Chapter 12]):

$$\frac{1}{|r_1\omega_1 - r_2\omega_2|} = \sum_{n=0}^\infty \frac{r_1^n}{r_2^{n+1}} \frac{4\pi}{2n+1} \sum_{m=-n}^n Y_{nm}(\omega_1)\overline{Y_{nm}(\omega_2)}.$$

Absolute convergence of the series (since $r_1 < r_2$) justifies interchanging integration and summation. Since $\eta = \mu - \mu_0$ has zero spherical average on every sphere, its $n = 0$ spherical-harmonic coefficient vanishes. Thus only the modes $n \ge 1$ contribute below.

$$F_s(\eta,\eta) = \sum_{n=1}^\infty \frac{4\pi}{2n+1} \sum_{m=-n}^n I_{n,m},$$

where

$$I_{n,m} = \int_0^\infty \int_{r_1}^\infty (r_1^s + r_2^s) \frac{r_1^n}{r_2^{n+1}} u_{n,m}(r_1)\overline{u_{n,m}(r_2)} \,\mathrm{d}r_2 \,\mathrm{d}r_1$$

and

$$u_{n,m}(r) = r^2 \int_{S^2} Y_{nm}(\omega)\eta(r\omega) \,\mathrm{d}\omega.$$

We need to analyze the kernel

$$(r_1^s + r_2^s)\frac{r_1^n}{r_2^{n+1}} = \frac{r_1^{s+n}}{r_2^{n+1}} + \frac{r_1^n}{r_2^{n+1-s}}.$$

Now we make an exponential change of variables. Put $e^p = r_1$ and $e^q = r_2$ with $p < q$ since $r_1 < r_2$. Then the kernel above becomes

$$e^{p(s+n)-q(n+1)} + e^{pn-q(n+1-s)}.$$

We factor out $e^{(s-1)/2(p+q)}$. Then, with $c_1 = (1+s)/2 + n$, $c_2 = (1-s)/2 + n$, and $\alpha := (s+1)/2$:

$$e^{-(p+q)}e^{\alpha(p+q)}\big[e^{c_1(p-q)} + e^{c_2(p-q)}\big].$$

Since $p < q$, the kernel is now of the form

$$e^{-(p+q)}e^{\alpha(p+q)}\big[e^{-c_1|p-q|} + e^{-c_2|p-q|}\big].$$

With $v_{n,m}(\xi) := u_{n,m}(e^\xi)e^{\alpha\xi}$, we now have

$$I_{n,m} = \int_0^\infty \int_{r_1}^\infty \big[e^{-c_1|p-q|} + e^{-c_2|p-q|}\big] v_{n,m}(p)\overline{v_{n,m}(q)}(e^{-(p+q)})\,\mathrm{d}r_2\,\mathrm{d}r_1.$$

Since $\mathrm{d}r_1 = e^p\mathrm{d}p$ as well as $\mathrm{d}r_2 = e^q\mathrm{d}q$, and since the expression is symmetric in p and q and the resulting sum is real, we conclude

$$\begin{aligned}
F_s(\eta,\eta) &= \sum_{n=1}^\infty \frac{4\pi}{2n+1}\sum_{m=-n}^n I_{n,m} \\
&= \sum_{n=1}^\infty \frac{4\pi}{2n+1}\sum_{m=-n}^n \frac{1}{2}\iint_{\mathbb{R}\times\mathbb{R}} \big[e^{-c_1|p-q|} + e^{-c_2|p-q|}\big] v_{n,m}(p)\overline{v_{n,m}(q)}\,\mathrm{d}p\,\mathrm{d}q \\
&= \sum_{n=1}^\infty \frac{4\pi}{2n+1}\sum_{m=-n}^n \frac{1}{2}\iint_{\mathbb{R}\times\mathbb{R}} K_n(p-q) v_{n,m}(p)\overline{v_{n,m}(q)}\,\mathrm{d}p\,\mathrm{d}q.
\end{aligned}$$

With $K_n(t) = e^{-c_1|t|} + e^{-c_2|t|}$. Note that we can rewrite the double integral as a convolution

$$\begin{aligned}
F_s(\eta,\eta) &= \sum_{n=1}^\infty \frac{4\pi}{2n+1}\sum_{m=-n}^n \frac{1}{2}\int_{\mathbb{R}} (K_n * v_{n,m})(q)\overline{v_{n,m}(q)}\,\mathrm{d}q \\
&= \sum_{n=1}^\infty \frac{4\pi}{2n+1}\sum_{m=-n}^n \frac{1}{2}\langle K_n * v_{n,m}, v_{n,m}\rangle_{L^2(\mathbb{R})}.
\end{aligned}$$

This allows us to apply Plancherel's and Convolution theorems whenever $c_j(s,n) > 0$ for $j = 1, 2$. Note that

- $c_1(s,n) = \dfrac{(1+s)}{2} + n \geq n+1$ for $s \geq 1$, hence positive for all $n \geq 1$;
- $c_2(s,n) = \dfrac{(1-s)}{2} + n \geq \dfrac{3-s}{2}$ is positive for $s < 3$.

For $1 \leq s < 3$, we obtain

$$\langle K_n * v_{n,m}, v_{n,m} \rangle = \langle \hat{K}_n \hat{v}_{n,m}, \hat{v}_{n,m} \rangle = \int_{\mathbb{R}} \hat{K}_n(\xi) |\hat{v}_{n,m}(\xi)|^2 \, \mathrm{d}\xi, \tag{6.5}$$

where $\hat{K}_n$ is the Fourier transform of the kernel $K_n(t) = e^{-c_1|t|} + e^{-c_2|t|}$:

$$\hat{K}_n(\xi) = \sqrt{\frac{2}{\pi}} \left(\frac{c_1}{c_1^2 + \xi^2} + \frac{c_2}{c_2^2 + \xi^2} \right).$$

Hence the kernel is positive-definite for $1 \leq s < 3$. For $1 \leq s < 3$, this gives

$$F_s(\eta, \eta) = \sum_{n=1}^{\infty} \frac{4\pi}{2n+1} \sum_{m=-n}^{n} \frac{1}{2} \int_{\mathbb{R}} \hat{K}_n(\xi) |\hat{v}_{n,m}(\xi)|^2 \, \mathrm{d}\xi \geq 0.$$

At $s = 3$, the terms $n \geq 2$ are treated by the same Fourier-transform argument, while for $n = 1$, one has $K_1(t) = e^{-3|t|} + 1$. Thus the $n = 1$ contribution is

$$\sum_{m=-1}^{1} \left(\sqrt{\frac{2}{\pi}} \int_{\mathbb{R}} \frac{3}{9 + \xi^2} |\hat{v}_{1,m}(\xi)|^2 \, \mathrm{d}\xi + \left| \int_{\mathbb{R}} v_{1,m}(p) \, \mathrm{d}p \right|^2 \right) \geq 0,$$

and therefore $F_3(\eta, \eta) \geq 0$ as well. This proves the claim.

6.3 Why This Method Breaks Down for $s > 3$

Now we show that this symmetrization, although it holds for $1 \leq s \leq 3$, breaks down in general. The goal is, for $s > 3$ and some radially symmetric measure ρ, to construct a (signed) measure η s.t. $F_s(\rho, \rho) > F_s(\rho+\eta, \rho+\eta)$, i.e. $F_s(\eta, \eta) < 0$, where $\rho, \rho + \varepsilon\eta \in D(\beta_s)$ for some $\varepsilon > 0$ small enough. Moreover, we choose ρ and η to be absolutely continuous with respect to the Lebesgue measure.

Recall that the bilinear form $F_s(\eta, \eta)$ can be written as

$$F_s(\eta, \eta) = \sum_{n=0}^{\infty} \frac{4\pi}{2n+1} \sum_{m=-n}^{n} \frac{1}{2} \iint_{\mathbb{R}\times\mathbb{R}} K_n(p-q) v_{n,m}(p) \overline{v_{n,m}(q)} \, \mathrm{d}p \, \mathrm{d}q, \tag{6.6}$$

where

$$K_n(t) = e^{-c_1(s,n)\,|t|} + e^{-c_2(s,n)\,|t|}, \qquad \begin{aligned} c_1(s,n) &= \frac{1+s}{2} + n, \\ c_2(s,n) &= \frac{1-s}{2} + n. \end{aligned}$$

and

$$v_{n,m}(\ln r) = r^{(s+5)/2} \int_{S^2} Y_{n,m}(\omega) \eta(r\omega) \, \mathrm{d}\omega. \tag{6.7}$$

In this notation, we used that our to-be-constructed measure η is absolutely continuous with respect to the Lebesgue measure. As a slight abuse of notation, we will now use $\eta(r\omega)$ as the notion of the density we construct. We will construct it using the $n = 1$ spherical harmonics, i.e. $\eta(r\omega) = \sum_{k=-1}^{1} f_k(r) Y_{1,k}(\omega)$ for some $f_k : [0, \infty) \to \mathbb{R}$ sufficiently regular. To avoid problems with ensuring that the image of η is real, we only take the $Y_{1,0}(\omega) = Y(\phi, \theta) = \sqrt{\frac{3}{4\pi}} \cos(\theta)$ harmonic. So the goal is to find a function $f : [0, \infty) \to \mathbb{R}$

such that $\eta(r\omega) := f(r)Y_{1,0}(\omega)$ defines a density for a measure $\rho + \varepsilon\eta \in D(\beta_s)$ for some appropriately chosen ρ and $\varepsilon > 0$.

If we insert $\eta(r\omega) := f(r)Y_{1,0}(\omega)$ into 6.7 and use the orthonormality of spherical harmonics, we obtain

$$v_{1,0}(\ln r) = r^{(s+5)/2} f(r), \qquad r > 0. \tag{6.8}$$

Hence it suffices to construct $v_{1,0} : \mathbb{R} \to \mathbb{R}$ and check its regularity. Then f is uniquely determined.

We need to ensure

$$\iint_{\mathbb{R}\times\mathbb{R}} \left[e^{-c_1(s,n)\,|p-q|} + e^{-c_2(s,n)\,|p-q|}\right] v_{1,0}(p) v_{1,0}(q)\,\mathrm{d}p\,\mathrm{d}q < 0,$$

which is possible with the following lemma.

Lemma 6.12. *Let $\alpha, \beta > 0$. Define, for $t \in \mathbb{R}$,*

$$K(t) := e^{-\alpha|t|} + e^{\beta|t|}$$
$$v_R(t) := 1_{(0,R)}(t) - 1_{(-R,0)}(t), \quad R > 0.$$

Then there exists $R > 0$ s.t.

$$\iint_{\mathbb{R}\times\mathbb{R}} K(p-q)\, v_R(p)\, v_R(q)\,\mathrm{d}p\,\mathrm{d}q < 0.$$

Proof. Note that the integrand is compactly supported. Put $G(q) := \int_{\mathbb{R}} K(p-q)\, v_R(p)\,\mathrm{d}p$. The integral above becomes

$$\int_{\mathbb{R}} G(q)\, v_R(q)\,\mathrm{d}q.$$

It is easy to check that $G(q) = -G(-q)$ for all $q \in \mathbb{R}$. Hence the integral simplifies to

$$2\int_0^R G(q)\,\mathrm{d}q.$$

It suffices to find an $R > 0$ s.t. $G(q) < 0$ for all $0 < q < R$. Note that

$$G(q) = \int_0^{-R} K(p-q)\,\mathrm{d}p \; + \int_0^R K(p-q)\,\mathrm{d}p.$$

One obtains for $R > \frac{1}{\beta}\ln 2$ and $0 < q < R$ that

$$\begin{aligned} G'(q) &= 2K(q) - K(R+q) - K(R-q) \\ &= 2e^{-\alpha q} + 2e^{\beta q} - e^{-\alpha(R+q)} - e^{\beta(R+q)} - e^{-\alpha(R-q)} - e^{\beta(R-q)} \\ &\le 2 + e^{\beta q}(2 - e^{\beta R}) \le 4 - e^{\beta R} \end{aligned}$$

which is negative for any fixed $R > \frac{1}{\beta}\ln 4 > 0$. Since $G(0) = 0$ and $G'(q) < 0$ for $0 < q < R$, the claim follows. □

With $\alpha = c_1(s,1)$ and $\beta = -c_2(s,1)$, we finally define $v_{1,0}(t) := 1_{(0,R)}(t) - 1_{(-R,0)}(t)$ for $R = -c_2(s,1)^{-1}\ln 4 + 1$. Then the negativity of the integral is ensured by the lemma. Using 6.8, we have that

$$f(r) = \frac{1}{r^{(s+5)/2}}\left[1_{(1,e^R)}(r) - 1_{(e^{-R},1)}(r)\right].$$

Note that $\operatorname{supp} f \subset [e^{-R}, e^R]$ and f is bounded. Since $Y_{1,0}$ is bounded, the density $\eta(r\omega) = f(r)Y_{1,0}$ defines a signed measure on $\mathbb{R}^3$ orthogonal to all radially symmetrical measures.

The last step is to choose the measure ρ and an $\varepsilon > 0$ such that $\rho + \varepsilon\eta \in D(\beta_s)$. We define ρ through its density $\rho(r\omega) = \frac{3}{4\pi(e^{3R}-e^{-3R})} 1_{(e^{-R},e^R)}(r)$. Then $\rho \in D(\beta_s)$ and is orthogonal to η. Using that $\sup_{\omega\in S^2} |Y_{1,0}(\omega)| = \sqrt{\frac{3}{4\pi}}$, we choose $\varepsilon = \frac{1}{2}\sqrt{\frac{3}{4\pi}}\frac{e^{-R(s+5)/2}}{e^{3R}-e^{-3R}}$.

Clearly, $1 = \int \mathrm{d}\rho = \int \mathrm{d}(\rho + \varepsilon\eta)$ and, by construction, $\rho + \varepsilon\eta$ is a positive measure since we choose

$$0 < \varepsilon < 2\varepsilon = \sqrt{\frac{3}{4\pi}}\frac{e^{-R(s+5)/2}}{e^{3R} - e^{-3R}} = \frac{\inf\{\rho(x) : x \in \operatorname{supp}\rho\}}{\sup\{|\eta(x)| : x \in \operatorname{supp}|\eta|\}}.$$

Moreover, since the densities are compactly supported and do not contain point masses, we have $\rho + \varepsilon\eta \in D(\beta_s)$.

Chapter 7

Summary

This thesis gave an overview of several important results regarding the ionization conjecture in the Schrödinger model. We translated the condition that a nucleus of charge Z can bind N electrons into a statement about spectra of Schrödinger operators, namely $E_{N,Z} < E_{N-1,Z}$, where $E_{N,Z} = \inf \sigma(H_{N,Z})$. In each chapter, we analyzed and refined necessary conditions given in terms of Z and N. Chapter 5 consists of original results that deepen the understanding of a crucial tool used in the analysis—$\alpha_{N,s}$. This explains several fundamental properties and limitations.

Another original result was the construction of a non-radial measure ρ that confirmed known limitations of the approach with restricting to radial measures when obtaining lower bounds on β_s.

The ionization conjecture in the formulation used in this thesis remains an important open problem. For example, if proven, it would improve rigorous analytical bounds on the ground state energies $E_{N,Z}$ in [4].

Appendix A

Technical Details

A.1 Details on Engel's Inequality/Titu's Lemma

In the proof of 5.13, we apply the following lemma with $x_1 = \sqrt{a^2+b^2}, x_2 = \sqrt{b^2+c^2}, x_3 = \sqrt{a^2+c^2}$ and $y_1 = d_{ab}$, $y_2 = d_{bc}, y_3 = d_{ac}$

Lemma A.1. *For positive numbers x_i, y_i, where $i \in \{1, 2, ..., n\}$, one has*

$$\frac{\left(\sum_i x_i\right)^2}{\sum_i y_i} \leq \sum_i \frac{x_i^2}{y_i}.$$

Proof. Apply CSI $\left(\sum_i a_i b_i\right)^2 \leq \left(\sum_i a_i^2\right)\left(\sum_i b_i^2\right)$ with $a_i = \frac{x_i}{\sqrt{y_i}}$ and $b_i = \sqrt{y_i}$. □

A.2 Geometric Properties

Proof of (5.4). We prove for all $x, y, z \in \mathbb{R}^3$ that $|x-y|+|y-z|+|x-z| \leq 3\sqrt{|x|^2+|y|^2+|z|^2}$. Consider $|x \pm y|^2 = x^2 + y^2 \pm 2\langle x, y\rangle$. Hence

$$|x-y|^2 + |y-z|^2 + |x-z|^2 = 2(|x|^2+|y|^2+|z|^2) - 2(\langle x,y\rangle + \langle y,z\rangle + \langle x,z\rangle)$$

and

$$|x+y+z|^2 = |x|^2+|y|^2+|z|^2 + 2(\langle x,y\rangle + \langle y,z\rangle + \langle x,z\rangle).$$

We conclude

$$|x-y|^2 + |y-z|^2 + |x-z|^2 = -|x+y+z|^2 + 3(|x|^2+|y|^2+|z|^2) \leq 3(|x|^2+|y|^2+|z|^2).$$

Applying the Cauchy-Schwarz inequality yields the claim:

$$|x-y| + |y-z| + |x-z| \leq \sqrt{3}\sqrt{|x-y|^2+|y-z|^2+|x-z|^2} \leq 3\sqrt{|x|^2+|y|^2+|z|^2}.$$

□

Proof of (5.5). Note that $(a^2+b^2)(b^2+c^2) = b^4+b^2(a^2+c^2)+(ac)^2 \geq b^4+b^2(2ac)+(ac)^2 = (b^2+ac)^2$. This also holds for all such mixed terms. Now, we apply the inequality to

$$\begin{aligned}
&\left(\sqrt{a^2+b^2}+\sqrt{b^2+c^2}+\sqrt{a^2+c^2}\right)^2 \\
&= 2(a^2+b^2+c^2) + 2\left(\sqrt{(a^2+b^2)(b^2+c^2)} + \cdots \text{other mixed terms}\right) \\
&\geq 2(a^2+b^2+c^2) + 2(a^2+b^2+c^2+ab+bc+ac) \\
&= 4S_2 + 2(ab+bc+ac), \qquad \text{where } S_j = a^j+b^j+c^j \\
&= 3S_2 + 1 \qquad \text{since } 1 = S_1^2 = S_2 + 2(ab+bc+ac).
\end{aligned}$$

□

A.3 Special Case of the IMS Localization Formula

Let $\xi \in W^{1,\infty}(\mathbb{R}^{3N}, \mathbb{R})$ with $\psi = \psi_{N,Z}$. We interpret $-\Delta_1$ in the quadratic form sense. Consider

$$\begin{aligned}\left\langle \xi^2\psi, (-\Delta_1)\psi \right\rangle &= \int_{\mathbb{R}^{3N}} \nabla_1(\xi^2\,\overline{\psi}) \cdot \nabla_1\psi = \int_{\mathbb{R}^{3N}} \xi^2\,\nabla_1\overline{\psi} \cdot \nabla_1\psi \;+ 2\int_{\mathbb{R}^{3N}} \xi\,\overline{\psi}\,\nabla_1\xi \cdot \nabla_1\psi \\ &= \int_{\mathbb{R}^{3N}} \xi^2|\nabla_1\psi|^2 + 2\int_{\mathbb{R}^{3N}} \xi\,\overline{\psi}\,\nabla_1\xi \cdot \nabla_1\psi,\end{aligned}$$

as well as

$$\begin{aligned}\langle \xi\psi, -\Delta_1(\xi\psi)\rangle &= \int_{\mathbb{R}^{3N}} |\nabla_1(\xi\psi)|^2 = \int_{\mathbb{R}^{3N}} |\psi\nabla_1\xi + \xi\nabla_1\psi|^2 \\ &= \int_{\mathbb{R}^{3N}} |\nabla_1\xi|^2|\psi|^2 + \int_{\mathbb{R}^{3N}} \xi^2|\nabla_1\psi|^2 + 2\,\mathrm{Re}\int_{\mathbb{R}^{3N}} \xi\,\overline{\psi}\,\nabla_1\xi \cdot \nabla_1\psi \\ &= \left\langle \psi, |\nabla_1\xi|^2\psi \right\rangle + \mathrm{Re}\left\langle \xi^2\psi, (-\Delta_1)\psi \right\rangle.\end{aligned}$$

This yields the desired identity:

$$\mathrm{Re}\left\langle \xi^2\psi, (-\Delta_1)\psi \right\rangle = \langle \xi\psi, -\Delta_1(\xi\psi)\rangle - \left\langle \psi, |\nabla_1\xi|^2\psi \right\rangle.$$

A.4 Details of Proposition 5.8

We provide a complete proof of Proposition 5.8. It requires a small lemma.

Lemma A.2. *If $x \neq y$, then*

$$\partial_\varepsilon|(1-\varepsilon)x - y| = \frac{(\varepsilon-1)|x|^2 + \langle x, y\rangle}{|(1-\varepsilon)x - y|}, \quad \varepsilon \in [0, \delta).$$

Proof. For a differentiable function $f > 0$, one has $(\sqrt{f(\varepsilon)})' = \frac{f'(\varepsilon)}{2\sqrt{f(\varepsilon)}}$ and thus for

$$\begin{aligned}f(\varepsilon) = |(1-\varepsilon)x - y|^2 &= \langle (1-\varepsilon)x - y, (1-\varepsilon)x - y\rangle \\ &= (1-\varepsilon)^2|x|^2 + |y|^2 - 2(1-\varepsilon)\langle x, y\rangle\end{aligned}$$

we have

$$\partial_\varepsilon|(1-\varepsilon)x - y| = (\sqrt{f(\varepsilon)})' = \frac{f'(\varepsilon)}{2\sqrt{f(\varepsilon)}} = \frac{f'(\varepsilon)}{2|(1-\varepsilon)x - y|} = \frac{(\varepsilon-1)|x|^2 + \langle x, y\rangle}{|(1-\varepsilon)x - y|}.$$

□

Proof of Proposition 5.8. Differentiating with help of Lemma A.2,

$$\begin{aligned}\partial_\varepsilon\mathcal{F}_s(\varepsilon) &= \sum_{1<j\leq N} \partial_\varepsilon \frac{(1-\varepsilon)^s|x_1|^s + |x_j|^s}{|(1-\varepsilon)x_1 - x_j|} = \sum_{1<j\leq N} \partial_\varepsilon \frac{(1-\varepsilon)^s + 1}{|(1-\varepsilon)x_1 - x_j|} \\ &= \sum_{1<j\leq N} \frac{s(\varepsilon-1)|(1-\varepsilon)x_1 - x_j| - ((1-\varepsilon)^s + 1)\frac{(\varepsilon-1)+\langle x_1, x_j\rangle}{|(1-\varepsilon)x_1 - x_j|}}{|(1-\varepsilon)x_1 - x_j|^2}.\end{aligned}$$

Thus

$$\begin{aligned}\partial_\varepsilon \mathcal{F}_s(0) &= \sum_{1<j\le N} \frac{1}{|x_1 - x_j|}\Big(-s - 2\frac{-1 + \langle x_1, x_j\rangle}{|x_1 - x_j|^2}\Big) \\ &= \sum_{1<j\le N} \frac{1}{|x_1 - x_j|}\big(-s - 2(-\tfrac{1}{2})\big) = a_1(1-s).\end{aligned}$$

We also have

$$\partial_\varepsilon \mathcal{G}_s(\varepsilon) = \partial_\varepsilon (1-\varepsilon)^{s-1} = 1 - s = \partial_\varepsilon \mathcal{G}_s(0), \qquad \mathcal{G}_s(0) = N.$$

As well as

$$\begin{aligned}\partial_\varepsilon(\mathcal{F}_s/\mathcal{G}_s)(0) &= \frac{\partial_\varepsilon \mathcal{F}_s(0)\cdot \mathcal{G}_s(0) - \mathcal{F}_s(0)\cdot \partial_\varepsilon \mathcal{G}_s(0)}{(\cdots)^2} < 0 \\ &\iff \partial_\varepsilon \mathcal{F}_s(0)\cdot \mathcal{G}_s(0) - \mathcal{F}_s(0)\cdot \partial_\varepsilon \mathcal{G}_s(0) < 0.\end{aligned}$$

With the calculations above,

$$\begin{aligned}&\partial_\varepsilon \underbrace{\mathcal{F}_s(0)}_{a_1(1-s)}\cdot \underbrace{\mathcal{G}_s(0)}_{N} - \underbrace{\mathcal{F}_s(0)}_{2\mathbb{T}_N}\cdot \underbrace{\partial_\varepsilon \mathcal{G}_s(0)}_{1-s} \\ &\qquad = \big(-N a_1 + \sum_{k=1}^{N} a_k\big)(s-1),\end{aligned}$$

which is strictly less than zero if and only if $a_1 > \frac{1}{N}\sum_{k=1}^N a_k$. This happens if and only if there exist j, k such that $a_k \neq a_j$ (since we can relabel the points so that $j = 1$). □

A.5 Explicit Constant in an Inequality Due to Lieb

Lemma A.3 (Explicit Constant in an Inequality Due to Lieb)**.** *For any $p \in [1,2]$, the constant C_p in (4.5) is given by*

$$C_p = \frac{3\sqrt{\pi}}{4}\frac{(4\pi)^{-p/3}}{p^{1+p/2}}\frac{\left(\frac{15\sqrt{\pi}}{8}\frac{\Gamma(3/p)}{\Gamma(7/2+3/p)}\right)^{p/2}\frac{\Gamma(3/p+1)}{\Gamma(3/p+7/2)}}{\left(\frac{\sqrt{\pi}}{4}\frac{\Gamma(3/p+1)}{\Gamma(3/p+5/2)}\right)^{1+5p/6}}.$$

Here Γ denotes the Gamma function,

$$\Gamma(r) := \int_0^\infty x^{r-1}e^{-x}\,dx.$$

Proof sketch. Following [6] and [5], it suffices to evaluate the three integrals

$$\int_{\mathbb{R}^3} f_p(x)\,dx, \qquad \int_{\mathbb{R}^3} |x|^p f_p(x)\,dx, \qquad \int_{\mathbb{R}^3} f_p(x)^{5/3}\,dx,$$

where

$$f_p(x) = \begin{cases}(1-|x|^p)^{3/2}, & |x|\le 1,\\ 0, & \text{otherwise.}\end{cases}$$

Using the spherical coordinates and evaluating the resulting Beta function in terms of the Gamma function $\frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)} = B(a,b) := \int_0^1 t^{a-1}(1-t)^{b-1}$, we obtain for the first integral

$$4\pi \int_0^1 r^2(1-r^p)^{3/2}\,dr = \frac{\sqrt{\pi}}{4}\frac{\Gamma\left(\frac{3}{p}+1\right)}{\Gamma\left(\frac{3}{p}+\frac{5}{2}\right)}.$$

For the remaining two integrals, the substitution $u = r^p$ yields

$$\frac{4\pi}{p}\int_0^1 u^{3/p}(1-u)^{3/2}\,du = \frac{3\sqrt{\pi}}{4p}\frac{\Gamma\left(\frac{3}{p}+1\right)}{\Gamma\left(\frac{3}{p}+\frac{7}{2}\right)},$$

and

$$\frac{4\pi}{p}\int_0^1 u^{(3-p)/p}(1-u)^{5/2}\,du = \frac{15\sqrt{\pi}}{8p}\frac{\Gamma\left(\frac{3}{p}\right)}{\Gamma\left(\frac{3}{p}+\frac{7}{2}\right)}.$$

Combining these identities gives the desired constant. □

A.6 Details on the Mean-Field Type Bounds

In this section, we provide a full proof [5] of Lemma 6.3—the first lower bound on $\alpha_{N,s}$ in terms of β_s. Namely, for every $N \geq 2, r > 0$, and $s \geq 0$,

$$\frac{(1+r)^{s+1}-(1-r)^{s+1}}{2r(s+1)}N\beta_s \leq (1+r)^s\left(\alpha_{N,s}(N-1)+\frac{1}{r}\right). \tag{A.1}$$

The proof goes by constructing specific measures. To make sure that those measures are in $H^{-1}(\mathbb{R}^3)$, we prove the following lemma from [5]:

Lemma A.4. *Let $r > 0, x \in \mathbb{R}^3 \setminus \{0\}$ and μ be a measure defined by*

$$\int_{\mathbb{R}^3} f\,\mathrm{d}\mu := \frac{1}{4\pi}\int_{S^2} f(x + r|x|\omega)\,\mathrm{d}\omega$$

for any measurable function f. Then $\mu \in H^{-1}(\mathbb{R}^3)$.

Proof. Let $r > 0, x_0 \in \mathbb{R}^3, \Omega := \{x \in \mathbb{R}^3 : |x - x_0| < r|x_0|\}$ and $\mathrm{tr}_{\partial\Omega} : H^1(\Omega) \to L^2(\partial\Omega)$ be the trace operator. Let $C > 0$ denote a generic constant. Assume $f \in H^1(\mathbb{R}^3)$, then

$$\left|\int_{\mathbb{R}^3} f\,\mathrm{d}\mu\right| = \left|\frac{1}{4\pi}\int_{S^2} f(x_0 + r|x_0|\omega)\,\mathrm{d}\omega\right| \leq C\left|\int_{\partial\Omega}(\mathrm{tr}_{\partial\Omega} f)(\omega)\,\mathrm{d}\omega\right| \leq C\|\mathrm{tr}_{\partial\Omega} f\|_{L^2(\partial\Omega)},$$

where we used the Cauchy-Schwarz inequality in the last step. The trace theorem shows

$$\|\mathrm{tr}_{\partial\Omega} f\|_{L^2(\partial\Omega)} \leq C\|f\|_{H^1(\Omega)} < \infty.$$

□

Proof of 6.3, [5]. Now, fix some $r > 0$ and $(x_1, ..., x_N) \in \mathcal{D}_N := \{X = (x_1, \dots, x_N) \in \mathbb{R}^3 : x_i \neq x_j \text{ for } i \neq j\}$. Define

$$\int_{\mathbb{R}^3} f \, \mathrm{d}\mu_j := \frac{1}{4\pi} \int_{S^2} f(x_j + r|x_j|\omega) \, \mathrm{d}\omega, \qquad j = 1, ..., N \tag{A.2}$$

$$\mu := \frac{1}{N} \sum_{j=1}^{N} \mu_j. \tag{A.3}$$

Clearly, these are probability measures. Moreover, by Lemma A.4, we have $\mu_j \in H^{-1}(\mathbb{R}^3)$, hence $\mu, \mu_j \in D(\beta_s)$ for all j. Using the definition and separating diagonal terms, we obtain

$$\begin{aligned} N^2 \iint_{\mathbb{R}^3 \times \mathbb{R}^3} \frac{|x|^s + |y|^s}{2|x-y|} \, \mathrm{d}\mu(x) \, \mathrm{d}\mu(y) &= \sum_{j,k=1}^{N} \iint_{\mathbb{R}^3 \times \mathbb{R}^3} \frac{|x|^s + |y|^s}{2|x-y|} \, \mathrm{d}\mu_j(x) \, \mathrm{d}\mu_k(y) \\ &= \sum_{j \neq k}^{N} \iint_{\mathbb{R}^3 \times \mathbb{R}^3} \frac{|x|^s + |y|^s}{2|x-y|} \, \mathrm{d}\mu_j(x) \, \mathrm{d}\mu_k(y) \\ &+ \sum_{j=1}^{N} \iint_{\mathbb{R}^3 \times \mathbb{R}^3} \frac{|x|^s + |y|^s}{2|x-y|} \, \mathrm{d}\mu_j(x) \, \mathrm{d}\mu_j(y). \end{aligned}$$

By construction, for all $x \in \operatorname{supp}(\mu_j)$, $|x - x_j| = r|x_j|$. Hence

$$|x| \leq |x - x_j| + |x_j| = (1 + r) \, |x_j|. \tag{A.4}$$

We start with the diagonal terms:

$$\begin{aligned} \iint_{\mathbb{R}^3 \times \mathbb{R}^3} \frac{|x|^s + |y|^s}{2|x-y|} \, \mathrm{d}\mu_j(x) \, \mathrm{d}\mu_j(y) &= \int_{\mathbb{R}^3} |x|^s \int_{\mathbb{R}^3} \frac{1}{|x-y|} \, \mathrm{d}\mu_j(x) \, \mathrm{d}\mu_j(y) \\ &\leq (1+r)^s |x_j|^s \int_{S^2} \int_{S^2} \frac{1}{|r|x_j|\eta - r|x_j|\omega|} \frac{\mathrm{d}\omega}{4\pi} \frac{\mathrm{d}\omega}{4\pi} \\ &= \frac{(1+r)^s}{r} |x_j|^{s-1}, \end{aligned}$$

hence

$$\sum_{j=1}^{N} \iint_{\mathbb{R}^3 \times \mathbb{R}^3} \frac{|x|^s + |y|^s}{2|x-y|} \, \mathrm{d}\mu_j(x) \, \mathrm{d}\mu_j(y) \leq \frac{(1+r)^s}{r} \sum_{j=1}^{N} |x_j|^{s-1}. \tag{A.5}$$

Similarly, we bound the off-diagonal terms:

$$\begin{aligned} \iint_{\mathbb{R}^3 \times \mathbb{R}^3} \frac{|x|^s + |y|^s}{2|x-y|} \, \mathrm{d}\mu_j(x) \, \mathrm{d}\mu_k(y) &\leq (1+r)^s |x_j|^s \int_{\mathbb{R}^3} \int_{S^2} \frac{1}{|x - x_k - r|x_k|\omega|} \frac{\mathrm{d}\omega}{4\pi} \, \mathrm{d}\mu_j(x) \\ &= (1+r)^s |x_j|^s \int_{\mathbb{R}^3} \frac{1}{\max\{|x - x_k|, r|x_k|\}} \, \mathrm{d}\mu_j(x) \\ &\leq (1+r)^s |x_j|^s \int_{\mathbb{R}^3} \frac{1}{|x - x_k|} \, \mathrm{d}\mu_j(x) \\ &\leq (1+r)^s \frac{|x_j|^s}{|x_j - x_k|}, \end{aligned}$$

where we used Newton's theorem multiple times. Thus

$$\sum_{\substack{j,k=1\\ j\neq k}}^{N} \iint_{\mathbb{R}^3\times\mathbb{R}^3} \frac{|x|^s+|y|^s}{2|x-y|}\,\mathrm{d}\mu_j(x)\,\mathrm{d}\mu_k(y) \leq (1+r)^s \sum_{j<k} \frac{|x_j|^s+|x_k|^s}{|x_j-x_k|}. \tag{A.6}$$

Combining (A.5) and (A.6) shows

$$\iint_{\mathbb{R}^3\times\mathbb{R}^3} \frac{|x|^s+|y|^s}{2|x-y|}\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(y) \leq (1+r)^s \left[\frac{1}{r}\sum_{j=1}^{N}|x_j|^{s-1} + \sum_{j<k}\frac{|x_j|^s+|x_k|^s}{|x_j-x_k|}\right]. \tag{A.7}$$

Using Lemma A.3 from [5], we obtain

$$N\int_{\mathbb{R}^3}|x|^{s-1}\,\mathrm{d}\mu(x) = \frac{(1+r)^{s+1}-(1-r)^{s+1}}{2r(s+1)}\sum_{j=1}^{N}|x_j|^{s-1}. \tag{A.8}$$

With the definition of β_s, one has

$$N^2\left[\int_{\mathbb{R}^3}|x|^{s-1}\,\mathrm{d}\mu(x)\right]\beta_s \leq N^2\iint_{\mathbb{R}^3\times\mathbb{R}^3}\frac{|x|^s+|y|^s}{2|x-y|}\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(y),$$

which together with (A.7) and (A.8) implies

$$\frac{(1+r)^{s+1}-(1-r)^{s+1}}{2r(s+1)(1+r)^s}N\beta_s \leq \frac{1}{r} + \frac{\sum_{j<k}\frac{|x_j|^s+|x_k|^s}{|x_j-x_k|}}{\sum_{j=1}^{N}|x_j|^{s-1}}.$$

Taking the infimum over $(x_1,...,x_N)\in\mathcal{D}_N$ yields the claim. □

Remark A.5. The proofs of refined bounds follow a similar strategy and use the same construction of measures μ,μ_j but improve the estimates of the weighted Coulomb terms using a convexity argument and their expansion in Legendre polynomials. Those methods take the angular structure into account which significantly sharpens the bounds.

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Statement on the Use of Auxiliary Means

In the preparation of this thesis, I used an AI model to suggest an exponential change of variables for the kernel $(r_1^s + r_2^s) r_1^n / r_2^{n+1}$ in an alternative proof of Lemma 6.10 (originally stated and proven in [5]). The initial idea of the proof, formalization, and the verification of the result were carried out independently by me. It is not claimed as a new contribution of this thesis. The original work in Section 6.3 (the construction of a non-radial measure for $s > 3$) was developed without the use of such tools.

Declaration of Originality

I truthfully declare that I have written the thesis independently, that I have fully and accurately cited all sources and aids used and have indicated everything that has been taken from the work of others unchanged or with modifications and that I have observed the KIT statutes for safeguarding good scientific practice in the currently valid version.

Place, Date: _____

_____

Zusatz zur Erklärung über die Aufbewahrung meiner Abschlussarbeit

Mir ist bekannt, dass meine Abschlussarbeit nach Ablauf der gesetzlichen Aufbewahrungsfrist von fünf Jahren gemäß den datenschutzrechtlichen Bestimmungen vernichtet werden kann. Es ist mein ausdrücklicher Wunsch und ich erkläre hiermit mein Einverständnis, dass meine Abschlussarbeit auch über diesen Zeitraum hinaus aufbewahrt und nicht vernichtet wird.

Place, Date: _____

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